Revisiting the Internationalisation–Performance Relationship in an Emerging Market: The Roles of Technological Capability and Foundational Digital Adoption

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Abstract

Purpose — This study revisits the internationalisation–performance relationship in Vietnam across three survey waves, testing whether technological capability moderates export-intensity effects and tracing how a Tier-1 digital indicator evolves across institutional generations.

Design/methodology/approach — Three waves of World Bank Enterprise Survey (Vietnam 2009, 2015, 2023; N = 2,958) underpin wave-specific and pooled OLS models with HC1 standard errors, quadratic FSTS terms, and capability interactions. TCI (quality certification, foreign-licensed technology) is the primary construct; a Tier-1 website indicator serves as digital-presence control.

Findings — An inverted-U between export intensity and labour productivity holds across all three waves, with turning points clustered between 39% and 46% — a band that stays durable over 14 years. The nonlinearity traces to a participation-margin effect:

restricted to exporters, the quadratic term loses significance. TCI positively moderates productivity and the I–P curvature across waves, while the website indicator follows a proxy-obsolescence trajectory: positive in 2009, null in 2015, negatively interactive in 2023.

Originality/value — In this zero-inflated setting the inverted-U reflects a participation-barrier effect rather than intensity saturation, and pooled Tier-1 coefficients mask progressive proxy obsolescence. Because the turning-point cluster (39–46% FSTS) is institutionally embedded, the debate moves from 'does internationalisation help?' to 'under what institutional conditions does the threshold shift?'

Keywords: internationalisation–performance; emerging markets; technological capability; threshold durability; Vietnam; firm productivity.

JEL classification: F23 (multinational firms; international business); O33 (technological change: choices and consequences); D22 (firm behaviour: empirical analysis); L25 (firm performance); O53 (economywide country studies: Asia including Middle East).

Paper type: Research paper.

Highlights

The internationalisation–performance relationship in Vietnamese firms is robustly non-linear: the Lind–Mehlum test rejects monotonicity in all three waves (2009 $p = .006$, 2015 $p = .009$, 2023 $p = .013$) and in the pooled sample ($p < .001$), with turning points clustered between 39 and 46 % of direct-export intensity.

Technological capability is the primary capability construct: TCI_z is the within-wave standardised mean of quality certification and foreign-licensed technology. A single-item website-presence indicator (DAI_z, Tier 1 only) is retained as a baseline digital-presence control given the WBES Vietnam measurement constraint; the two share no items.

TCI_z is positively associated with productivity in all three waves ($\beta = 0.215$, 0.128, 0.123) and pooled ($\beta = 0.179$, $p < .001$), and moderates the curvature in three of four panels (M3 joint $p = .040$, .713, .027 and .003).

The inverted-U turning point is structurally durable across waves: range 39.3–46.2 % across 2009 / 2015 / 2023 (pooled 39.7 %), a ~7-percentage-point band sustained through WTO accession (2007), the Global Financial Crisis (2008–2009), CPTPP/EVFTA implementation (2018–2020), the COVID-19 disruption (2020–2022), and the diffusion of digital-payment infrastructure.

The inverted-U is fundamentally a step function across the export-participation barrier rather than a continuous within-exporter saturation curve: re-fitted on the exporter-only sub-sample (FSTS > 0; pooled N = 669), the quadratic FSTS term loses significance (β = -0.200, p = .660); only ~1.0 % of pooled firms sit within ±5 percentage-points of the turning point. The productivity-relevant friction in a transitional economy binds at the participation margin, not within-exporter intensity.

The baseline website-presence indicator is descriptively positive in 2009 (β = 0.175), null in 2015 (β = -0.044), and positive in 2023 (β = 0.095); the negative DAI × FSTS interaction observed only in 2023 (FSTS_c × DAI_z = -0.912, p = .043) is read as Tier-1 proxy obsolescence as website ownership diffuses to near-universal levels (49.8 % in 2023 vs 42.5 % in 2009), not as evidence on dynamic digital capability moderation.

1. Introduction

1.1 Background and motivation

Vietnam offers an analytically valuable setting for revisiting the internationalisation–performance (I–P) relationship because firms expand abroad under conditions of institutional transition, uneven capability accumulation, and rapidly changing digital infrastructure.

In such settings, internationalisation does not lend itself to the assumption of a simple linear performance premium. Foreign involvement opens access to larger markets, learning, and diversified revenue, yet it also brings rising coordination costs, heavier information-processing burdens, and organisational strain as that involvement deepens (Wright et al., 2005; Cuervo-Cazurra and Genc, 2008; Wu et al., 2016). This tension lies at the heart of the I–P literature. A long tradition of work holds that internationalisation improves performance at lower and intermediate levels through scale, learning, and diversification, while complexity and coordination burdens drag returns toward zero or below them at higher levels. Meta-analytic evidence indicates that this nonlinearity is a central feature of the relationship, not an empirical anomaly (Vernon, 1979; Lu and Beamish, 2004; Hennart, 2007; Coviello et al., 2017; Marano et al., 2016). Digitalisation layers a further complication onto this debate. Digital tools can cut communication frictions, speed transactions, and support cross-border coordination, but the benefits do not arrive automatically. Their realised value depends on whether a firm holds the organisational depth, absorptive capacity, and complementary routines needed to convert digital adoption into productivity gains (Cohen and Levinthal, 1990; Vial, 2019; Verhoef et al., 2021; Stallkamp and Schotter, 2021; Petricevic and Teece, 2019). Digital capability, then, is poorly described as a universally beneficial resource whose payoff stays constant across firms and over time. The Vietnamese setting makes this issue especially pointed. Firms here are embedded in an economy that is upgrading capability and passing through digital transition at the same time, so both the gains from internationalisation and the payoff from digitalisation may shift across stages of development. The central question for this study follows directly: does digital capability in Vietnam behave as a stable performance-enhancing asset, or as a stage-contingent resource whose value changes over the lifecycle of internationalisation? Three institutional turning points shape the observation window. Vietnam acceded to the World Trade Organization in early 2007, which opened the period preceding the 2009 wave and converted a domestically oriented exporter cohort into one with broader market exposure but limited absorptive infrastructure. The 2015 wave captures the middle of a second phase, in which expanding manufacturing exports coexisted with under-developed digital trade infrastructure, weak cross-border logistics integration, and an exporter cohort still concentrated in labourintensive segments. The 2023 wave follows the launch of the National Digital Transformation Programme in 2020, the rapid expansion of cross-border e-payment and e-commerce platforms, and the rebalancing of foreign direct investment toward digitally-mediated and services-linked production. The three waves therefore observe firms under structurally different combinations of internationalisation pressure and digital infrastructure availability, which is what makes the

lifecycle reading testable rather than purely conceptual. These shifts are not cosmetic. The exporter cohort itself changes composition across the three waves: the share of firms reporting any positive direct-export intensity falls as services and FDI-linked supply-chain firms enter the sample, while average foundational digital adoption rises with the diffusion of websites, electronic payment systems, and digital transaction interfaces. To read the I–P relationship and the digital-adoption channel as fixed structural facts across this fourteen-year window is to miss that the underlying firm population, the binding coordination costs, and the institutional scaffolding for cross-border trade all change materially.

A central empirical regularity that emerges from this design is that the inverted-U observed in the full sample is fundamentally driven by the participation margin — the productivity gap between firms that cross from non-exporting to any positive export intensity — rather than by saturation of coordination costs within the exporter subsample. Once participation is netted out, the within-exporter intensity curvature is statistically indistinguishable from flat. The Vietnamese pattern therefore reads as a *step function across the export-entry threshold* in a transitional economy, not as a continuous saturation curve in the meta-analytic sense documented in developed-market samples. We treat this step-function structure as a headline empirical contribution rather than as an ancillary robustness probe, because it locates the productivity-relevant friction at the participation margin where institutional and capability constraints bind hardest in a transitional setting.

1.2 Research gap

This study addresses three gaps that remain unresolved in the existing literature.

One gap concerns how digitalisation is theorised. Much existing work treats it as a broadly positive resource and pays little attention to temporal and contextual variation in its payoff. That treatment risks implying a relatively stable performance premium across firms and across stages of economic transition (Strange and Zucchella, 2017; Goldfarb and Tucker, 2019). The returns are more likely to be uneven, since firms differ in scale, routines, and complementary capability.

A second gap concerns construct clarity. The distinction between technological capability and digital adoption remains underdeveloped. Technological capability refers to deeper firm-internal stocks of learning, problem-solving, process improvement, and innovation capacity (Lall, 1992); foundational digital adoption reflects a more basic layer of digital readiness and digitally enabled interfaces or transaction mechanisms (Bharadwaj et al., 2013; Verhoef et al., 2021; Hanelt et al., 2021; Brouthers et al., 2016). The two are related, but treating them as interchangeable, and collapsing them into a single umbrella variable, blurs the mechanisms that link digitalisation to performance.

The remaining gap is methodological. Pooled estimates can obscure substantial lifecycle heterogeneity. Where firms and institutions pass through distinct stages of transition, as in Vietnam, the roles of internationalisation, technological capability, and digital adoption may vary considerably across survey waves. A pooled model identifies average effects well enough, but it falls short once the underlying relationships shift over time. A design that combines pooled and wave-specific analysis is therefore needed to determine whether the observed effects are stable or stage contingent.

Recent meta-analytic work reinforces this concern. Pisani, Garcia-Bernardo, and Heemskerk (2020, SMJ) find that the inverted-U relationship between internationalisation and performance weakens substantially — and sometimes disappears — under more rigorous identification in cross-national pooled samples, suggesting that aggregation across heterogeneous institutional contexts inflates apparent functional-form regularity. Wu, Fan, and Chen (2022, MIR) extend this argument by showing in a meta-analysis of emerging-market multinationals (EMNEs) that institutional context moderates I–P effect sizes more powerfully than firm-level capability variables. Taken together, these findings imply that within-country longitudinal designs — which hold institutional context constant while varying time — provide a more credible test of whether the inverted-U is an artefact of cross-national pooling or a genuine relationship that persists as the institutional environment evolves. Vietnam's three-wave WBES panel (2009, 2015, 2023) provides exactly this design. Vietnam's macroeconomic and trade profile at the close of the observation window corroborates the transitional setting: GDP growth reached 7.09% in 2024, FDI inflows were US$20.17 billion (4.23% of GDP), and the export structure shifted substantially toward machinery and electronics — at 45.8% of total trade value, displacing the historically dominant textiles cluster — reflecting a structural upgrade in production capability that proceeded in parallel with the three survey waves (World Bank, 2025c).

1.3 Contribution

This study makes four contributions to the literature.

It first refines the I–P debate. The Vietnamese evidence supports a nonlinear relationship, but the salience and visibility of that relationship vary across time. Nonlinearity, on this reading, is not a fixed structural fact that appears identically in every period; the I–P curve has to be read alongside the broader capability environment.

The study also improves construct validity by separating technological capability from foundational digital adoption. The distinction matters theoretically, because deeper capability stocks and basic digital enablement may drive performance through different channels, and empirically, because the two constructs do not trace identical patterns across the Vietnamese waves.

A further contribution is a cross-wave, stage-dependency reading of digital internationalisation. The WBES data are repeated cross-sections rather than a firm panel, so within-firm trajectories cannot be observed directly; even so, the cross-wave evidence points to digital capability as neither a universally stable premium nor a uniformly ineffective resource. It appears instead as an uneven, snapshot-contingent source of performance heterogeneity across institutional phases — which helps explain how pooled average effects coexist with substantial wave-specific differences.

The final contribution reframes the inverted-U itself: the Vietnamese evidence suggests that the robustly identified turning point (39–46 % FSTS) emerges as a *step function across the export-participation barrier* rather than as a continuous coordination-cost saturation curve. The decomposition is sharp: re-fitted on the exporter-only sub-sample, the quadratic FSTS term collapses to a non-significant $\beta = -0.200$ ($p = .660$), and only ~1.0 % of pooled firms occupy the ±5 percentage-point neighbourhood of the turning point. This locates the productivity-relevant friction at the participation margin — where institutional, capability, and sunk-cost barriers bind hardest in a transitional economy —

rather than within the intensity tail where conventional Uppsala-style coordination-cost arguments operate. The within-exporter flatness is consistent with the view that, once a Vietnamese firm has absorbed the fixed costs of crossing into international markets, additional intensity does not by itself trigger a coordination-cost saturation strong enough to reverse the productivity gain. This step-function reading qualifies the conventional inverted-U interpretation and points toward participation-margin frictions, rather than intensity-margin saturation, as the dominant binding mechanism in transitional-economy I–P dynamics. The divergence between this pattern and those documented in digitally advanced economies, where the coordination-cost mechanism is attenuated by comprehensive digital infrastructure, anchors the digitally transitional end of the institutional spectrum and motivates institution-level mechanisms as candidate moderators of I–P curve location.

1.4 Roadmap

The remainder of the paper is organised as follows. Section 2 develops the theoretical framework and hypotheses, and Section 3 describes the data, variables, and empirical strategy. Section 4 presents the results, Section 5 discusses the theoretical and managerial implications, and Section 6 concludes.

2. Theory and hypotheses

2.1 Internationalisation and firm performance

In a transitional economy such as Vietnam, the relationship between internationalisation and firm performance is unlikely to be linear. Foreign expansion can lift performance through greater market reach, diversification, and learning from external environments; firms use internationalisation to spread fixed costs, reach new customers, and acquire knowledge that supports operational improvement. Deeper international involvement, at the same time, tends to generate coordination costs and organisational burdens. A firm must manage multiple markets, reconcile diverse customer demands, process more information, and sustain tighter managerial control. As foreign expansion intensifies, these costs may outgrow the benefits, producing diminishing returns or even a decline in performance. This basic logic underpins the classic nonlinear view of the I–P relationship (Contractor, 2007; Hennart, 2011; Marano et al., 2016). Process accounts of internationalisation, including the updated Uppsala framework, similarly emphasise that performance gains and adjustment costs unfold incrementally as firms accumulate market knowledge and commitment (Vahlne and Johanson, 2017; Knight and Liesch, 2016).

This argument carries particular weight in Vietnam, where firms operate under uneven capability conditions. Some convert foreign expansion into learning and scale advantages; others meet organisational strain sooner. What we expect, then, is not a uniformly positive slope, but a nonlinear relationship in which the gains from internationalisation become progressively harder to sustain. Two opposing forces underpin the curvature.

On the upside, increasing direct-export in-

tensity creates scale economies, knowledge spillovers from foreign customers, and learning-byexporting effects that lift productivity (Wagner, 2007).

On the downside, coordinating pro-

duction for institutionally distant markets imposes information-processing costs that grow nonlinearly: each additional foreign market adds compliance demands, customer-relationship overhead, and supply-chain dependencies whose marginal coordination cost rises faster than the

marginal scale benefit beyond a threshold. In a transitional setting where ports, trade-finance institutions, digital marketplaces, and dispute-resolution mechanisms are still maturing, this threshold may bind at a lower level of export intensity than in mature economies, sharpening the curvature relative to the meta-analytic baseline (Hennart, 2007; Wagner, 2007; Marano et al., 2016). The mechanism is institutional transaction costs (Williamson, 1985; Hennart, 2007): in a transitional economy, enforcement gaps, logistics deficits, and information asymmetries raise the marginal cost of managing each additional foreign market, causing the post-threshold decline to bind at lower export intensity than in settings where digital infrastructure and contract-enforcement quality have already absorbed much of this friction. A second consideration arises from the structure of direct-export intensity in WBES Vietnam: the FSTS variable is bounded at zero and is heavily zero-inflated (the non-exporter share is 71.6 % in 2009, 79.3 % in 2015 and 81.2 % in 2023; pooled 77.4 %). The internationalisation– performance relationship therefore comprises two analytically distinct margins. The participation margin is the step from FSTS = 0 to FSTS > 0, where firms cross the entry threshold into international markets and absorb fixed entry costs. The intensity margin is the variation of FSTS within the exporter subsample.

Theory predicts that productivity gains can accrue at either

margin: the participation margin captures learning-by-exporting and exposure to foreign-market demand; the intensity margin captures scale economies on the upside and coordination burdens on the downside (Bernard et al., 2007; Wagner, 2007). In transitional economies with thin export infrastructure, the participation margin is plausibly the dominant productivity-enhancing margin, while the intensity margin may show diminishing or null returns once participation has been crossed.

H1. In Vietnam's zero-inflated export setting, when firms cross the participation barrier into international markets, the internationalisation–performance relationship is non-monotonic in the full sample and is best understood through a participation-and-intensity structure, because the coordination-cost and learning mechanisms that shape I–P returns operate differently at the participation margin versus within the exporter subsample (Hennart, 2007; Wagner, 2007). (H1a, participation margin) Crossing from non-exporting (FSTS = 0) to exporting (FSTS > 0) is positively associated with labour productivity, capturing the learning-and-scale jump at entry, until the intensity margin begins where scale benefits plateau and coordination costs grow. (H1b, intensity margin) Within the exporter subsample, additional direct-export intensity is expected to exhibit weaker, diminishing, or non-significant marginal returns relative to the participation margin.

The exporter-only specification (Panel H) provides the most direct test of H1b: if the inverted-U curvature reflects within-exporter intensity variation, the quadratic term should remain significant in that sub-sample. The near-flat pattern observed in Panel H (pooled $FSTS\_c^2$ $\beta = -0.200$, $p = .660$) instead indicates that the participation margin (H1a) is the dominant source of the full-sample curvature.

2.2 Foreign-technology and standards capability and firm performance

Following the Lall (1992) tradition for emerging-market firms, this paper uses a measurementtight reading of technological capability: a foreign-technology and standards capability that captures a firm's exposure to externally validated technological inputs — internationally recognised quality certification and foreign-licensed technology. This is one observable facet of the broader Cohen and Levinthal (1990) absorptive-capacity construct and the dynamic-capability construct of Teece (2007), but the present measure does not claim to identify the full absorptive-capacity stock; it captures the externally facing component of that stock (Eisenhardt and Martin, 2000; Karna et al., 2016).

In international settings, this externally facing capability is particularly

important because firms must respond to unfamiliar markets, meet foreign quality standards, and integrate externally licensed technology into organisational routines. A firm with stronger foreign-technology / standards capability is more likely to transform international exposure into productivity gains.

It can adjust products and processes to meet foreign requirements more effectively, integrate licensed foreign technology into production, and cope better with the operational demands created by export activity. Even when this capability

does not fundamentally alter the curvature of the I–P relationship, it should improve the firm's overall performance level by raising the productivity floor among exposed firms. This implies a positive direct association between foreign-technology / standards capability and firm performance in Vietnam. We treat it here as a construct that should raise the firm's capacity to benefit from internationalisation and should also support productivity more broadly, without assuming that it indexes the full innovation-and-R&D dimension. Operationally, the primary TCI_z is built from two items: internationally recognised quality certification (b8) and foreign-licensed technology (e6). These items measure exposure to foreigntechnology and standards channels rather than internal R&D effort or patent activity. A broader innovation-augmented composite (TCI_full, adding product innovation h1 and R&D activity h8) is reported in 4.5 Panel A as a boundary condition: if direct effects attenuate when innovation items are added, this indicates that the primary measure is informative specifically about the foreign-technology / standards channel, not the broader absorptive-capacity stock.

H2. In Vietnam's transitional export environment, when firms hold stronger foreign-technology and standards capability (TCI_z), labour productivity is higher, because exposure to internationally recognised quality certification and foreign-licensed technology raises a firm's capacity to meet foreign market requirements and integrate external technological inputs into production routines (Lall, 1992; Cohen and Levinthal, 1990), before the productivity premium from this capability channel is bounded by the firm's absorptive depth relative to the full innovation-and-R&D stock (Eisenhardt and Martin, 2000).

2.3 Website-based digital presence and firm performance

The primary DAI_z used in this paper is a website-based digital presence measure: a binary indicator of whether the firm has its own website. This is a foundational and cross-wave-comparable marker of digital adoption — it does not measure transaction-level digital integration, electronic payment infrastructure, or digital transformation in the Bharadwaj et al. (2013) / Verhoef et al. (2021) / Vial (2019) sense. We use it here precisely because it is the only digital indicator the WBES instrument carries comparably across the 2009 / 2015 / 2023 waves; transaction-level items such as electronic-payment shares appear only in the 2023 questionnaire (Brouthers et al., 2016; Goldfarb and Tucker, 2019). For firms participating in foreign markets, even foundational website adoption may be valuable. A website lowers the cost at which foreign customers can locate the firm, supports asynchronous communication across time zones, and signals basic legitimacy in international transactions. The scope of these gains is bounded — a website does not by itself integrate transactions, supply chains, or decision-making — and the realised payoff depends on whether the firm has the scale and routines to translate online visibility into export business. Website-based digital presence should therefore show a positive average association with performance, but not necessarily one that is uniform across waves and stages. The information value of a website may have shifted across the 2009–2023 window:

in 2009 a website was a

distinguishing market interface; by 2023 it is closer to a routine marker of basic digital presence. The 5 discussion takes this proxy obsolescence reading seriously as one alternative to a pure stage-contingency story. Following Verhoef et al. (2021), digital capability can be located on a four-tier hierarchy: Tier 1 — digital presence (websites, e-mail); Tier 2 — digital communication and basic e-commerce; Tier 3 — digital process integration (electronic payment, supply-chain digitisation); Tier 4 — dynamic digital capability (data-driven decision-making, AI integration). The primary DAI_z

anchors at Tier 1 only. A Tier 3-style extension (DAI_rich) is reported in 4.5 Panel B for the 2023 wave, where electronic-payment items become available. The label foundational website adoption therefore tracks what the construct can actually identify across the 2009–2023 window.

Because DAI_z is anchored at Tier 1 (website presence only), the construct is not comparable to digital-adoption measures that also capture Tier 2 transaction-enabling items such as electronic-payment intensity. In settings where Tier 2–3 digital infrastructure is already mature and widely accessible, basic digital adoption may interact with export intensity through a different mechanism — functioning as a conditional scaling complement rather than a coordination-strain amplifier — because the surrounding ecosystem can absorb cross-border transaction processing that a website alone cannot.

Given that DAI_z reduces to a single binary item — c22b (own-website presence) — it captures Tier-1 digital presence rather than dynamic digital capability. By 2023, website ownership has diffused to near half of the Vietnamese firm population (49.8 % in 2023 vs 42.5 % in 2009) and increasingly functions as an organisational hygiene factor rather than a marker of digital strategy. **Accordingly, DAI_z is not formalised as a primary hypothesis-bearing construct in this paper.** It is retained in all specifications as a baseline digital-presence control to absorb website-ownership variance, and its level association with productivity is reported descriptively. Any future test of digital capability moderation would require richer Tier 2–4 indicators (electronic-payment intensity, ERP integration, e-commerce platform adoption, AI-augmented operations) that are not available cross-wave in the WBES Vietnam instrument.

Note on hypothesis numbering. Because DAI_z is not advanced as a primary hypothesis in this study, no H3 is formulated in the present paper. The exploratory digital moderation probe that follows (Section 2.4) is designated H4 to preserve alignment with the broader dissertation framework, in which H3 conventionally corresponds to institutional moderation in multi-country papers (P5 China, P7 Multi-country). Readers expecting H1–H2–H3–H4 sequencing should note this deliberate gap and the rationale above.

WBES-scope caveat. Because the WBES Vietnam instrument carries only the website-presence binary (c22b) comparably across the 2009 / 2015 / 2023 waves, this paper does not attempt to test higher-tier digital-transformation mechanisms such as data-driven decision-making, AI integration, or ERP-based process digitisation. Any descriptive coefficient on DAI_z should be read as reflecting baseline digital-presence variation, not dynamic digital capability. This caveat shields the present analysis from over-claiming on the limited Tier-1 information that a single binary website indicator can provide.

2.4 Stage-contingent digital value

Because the WBES Vietnam instrument cannot identify Tier 2–4 digital capability across waves, the present paper does not advance a primary digital-transformation claim. The following discussion frames a *secondary, exploratory probe* of whether the Tier-1 website indicator interacts with export intensity differently across waves — read as a within-WBES sensitivity check on Tier-1 baseline behaviour, not as a hypothesis test of dynamic digital capability. The exploratory claim is that even baseline digital presence may be stage contingent in scope: In early phases of internationalisation, digital tools may create relatively direct gains by helping firms communicate faster, access markets more easily, and manage transactions more efficiently. In later phases, however, the benefits of digitalisation may become more conditional because firms face more complex coordination demands.

Under such conditions, the value of founda-

tional digital adoption depends increasingly on whether digital tools are embedded in broader organisational routines and aligned with export scale (Banalieva and Dhanaraj, 2019; Petricevic and Teece, 2019). This argument implies that digitalisation can be a double-edged sword in a transitional economy. It may generate observable gains, but those gains are uneven and dependent on timing, scale, and complementary capability. In some phases, digital adoption may operate mainly as a direct performance-enhancing factor. In others, it may weaken, disappear, or become conditional on the firm's level of internationalisation. This expectation is particularly relevant in Vietnam, where firms operate in an environment of transition rather than full institutional and capability stability. A lifecycle interpretation is therefore more appropriate than a uniform premium interpretation. Two further considerations sharpen the prediction. First, when the exporter cohort is concentrated in low-export-intensity manufacturing, foundational digital tools mainly perform a market-access role: they help the firm find customers, communicate prices and product information, and process simple transactions. The marginal productivity gain from this role is positive but broadly distributed across the export-intensity range.

Second, when the exporter cohort

shifts toward firms that operate at higher export intensity and engage in tighter cross-border coordination, the same Tier 1–2 digital tools begin to interact with the marginal coordination cost of additional foreign markets. Whether this interaction is substitutive (digital tools lower coordination cost and amplify the productivity dividend) or complementary with diminishing returns (digital tools at high intensity reveal the absence of deeper integration and amplify coordination strain) is fundamentally an empirical question that this paper treats as the test of H4.

Two parallel mechanisms behind any later-wave DAI compression. Any negative DAI × FSTS interaction observed in the later waves can be read through two complementary mechanisms that this paper treats as jointly operating rather than mutually exclusive. *Mechanism A — stage-dependent coordination complexity*: at high export intensity, the marginal coordination demand of additional cross-border transactions exceeds what a single Tier-1 website indicator can support; the binding friction is the absence of deeper Tier 2–3 process-integration, payment, and logistics digitisation rather than the website itself. *Mechanism B — construct obsolescence under population diffusion*: as website ownership diffuses toward the near-universal upper bound (49.8 % by 2023, vs 42.5 % in 2009), the c22b indicator increasingly functions as an organisational hygiene marker rather than a differentiating capability signal, so its measured association with productivity attenuates without any structural change in the underlying coordination mechanism. The two readings are not antagonistic: they correspond to different sources of the same empirical pattern — Mechanism A predicts compression conditional on export intensity within any wave where Tier-1 infrastructure is the binding constraint, while Mechanism B predicts compression across waves as the indicator loses cross-sectional information content. The 2009 / 2015 / 2023 design used in this paper cannot fully separate the two channels within the WBES Tier-1 instrument, and we treat both readings as legitimate within the limits of the available data. This framing matters for theoretical positioning: it locates DAI compression in a transitional-economy context where the construct is bounded by survey instrumentation rather than by the underlying digital-transformation phenomenon, and it pre-empts an interpretation of any null or sign-shifting result on H4 as evidence against dynamic digital capability moderation.

H4 (exploratory probe — not a primary hypothesis). The productivity relevance of baseline website-presence (DAI_z, Tier 1 only) may vary across phases of internationalisation and institutional transition. Any moderation of the export-intensity curve by baseline digital presence is therefore expected to be wave-specific rather than uniformly present across periods, with the strongest within-sample detectability anticipated in the 2023 wave. *This probe is reported descriptively; the available Tier-1 binary indicator cannot adjudicate broader digital-transformation claims, and a null or sign-shifting result on H4 should be read as construct-tier obsolescence rather than as evidence against dynamic digital capability moderation.*

Technological Capability (TCI_z) H2 H2 (mod.)

InternationalisationH1 (non-monotonic) Firm performance (FSTS_c, FSTS_c²) (ln labour productivity) H4 (mod., exploratory probe) DAI_z (baseline control)

Baseline website-presence (DAI_z, Tier 1 only)

Controls: lnEmp, FirmAge, ForeignOwned, sector FE [+ wave FE]

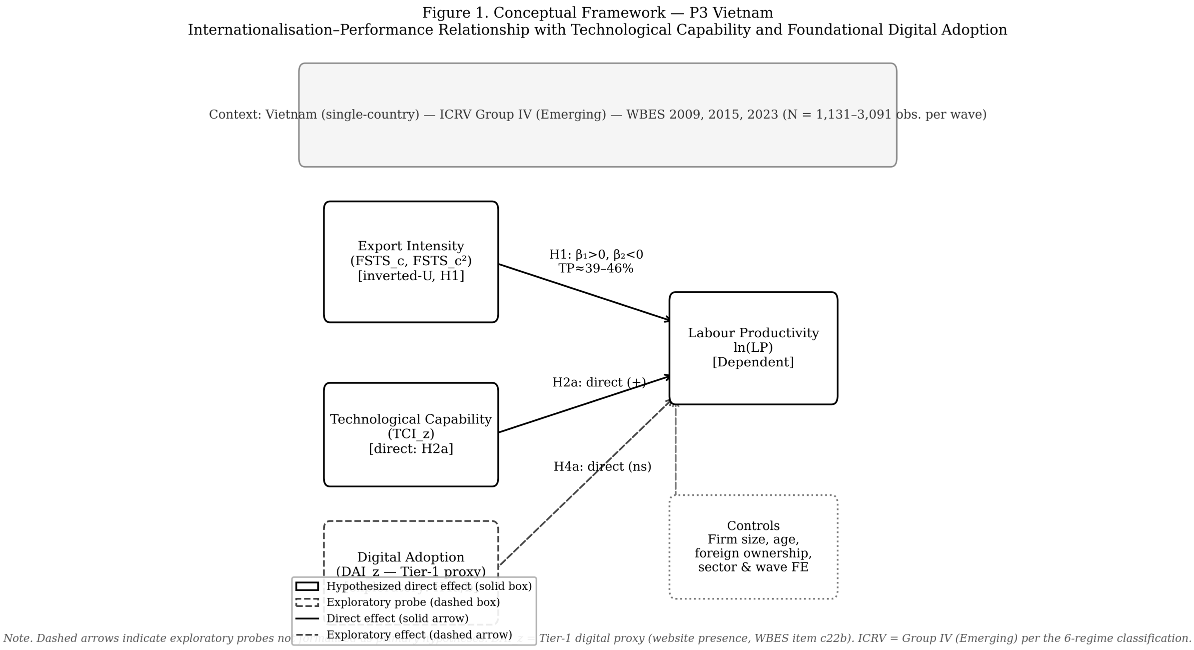

Figure 1: Figure 1: Conceptual model — TCI, DAI, and export intensity as determinants of firm performance (Vietnam, 3-wave)

Figure 1. Conceptual model for Paper 3 (Vietnam).

Note: Solid arrows represent hypothesised direct and curvilinear effects. Dashed arrows represent moderating effects. The independent variable — internationalisation intensity (FSTS, centred FSTS_c and FSTS_c²) — is placed at left; the dependent variable — firm performance, measured as ln(labour productivity) = ln(annual revenue PPP / permanent workers) — is placed at right, following left-to-right causal flow convention. H1 (inverted-U nonlinearity: $\beta_1 > 0$, $\beta_2 < 0$) is grounded in the Uppsala model (Johanson & Vahlne, 1977, 2009) and coordination-cost theory (Hitt et al., 1997). H2 specifies technological capability (TCI_z, Resource-Based View: Barney, 1991) as a level-shifting direct effect and secondary moderator on the FSTS–performance curve; instrument: industry-level TCI mean (causal IV identification). H4 treats website-based digital adoption (DAI_z, Tier-1 only) as an exploratory probe of the Digital Capability Lens (Banalieva & Dhanaraj, 2019); measurement constraint acknowledged — single binary indicator across 2009/2015/2023 waves limits inferential strength. (+) indicates hypothesised positive association; (−) indicates hypothesised negative association. Control variables (ln firm size, firm age, foreign ownership, broad-sector fixed effects, wave fixed effects) enter additively in all models but are omitted from the figure for clarity. ICRV context: Vietnam — Nhóm IV (Lower_mid_transition, ICRV Group 4 of 6; lower-middle-income transition economy; WGI Rule of Law ≈ −0.09 in 2023, World Bank). Data: WBES 2009, 2015, 2023; N = 2,958 (pooled), n_wave = 989/956/1,013.

3. Data, variables, and empirical strategy

3.1 Data structure

The empirical analysis uses harmonised firm-level evidence for Vietnam across three waves of the World Bank Enterprise Survey: 2009, 2015 and 2023 (World Bank, 2010, 2016, 2024). Estimating the models separately by wave makes it possible to observe whether relationships are stable or time specific, while pooled estimation identifies average effects across the broader period. The effective estimation sample varies across model specifications because of missing values in the capability variables. In the full wave-specific models, the usable samples are 989 observations for 2009, 956 for 2015, and 1,013 for 2023. The pooled full model contains 2,958 observations. This structure provides sufficient variation to compare direct effects and conditional patterns across stages.

3.2 Variables

Firm performance is measured by log labour productivity (lnLP). Internationalisation is measured by direct-export intensity (FSTS), mean-centred within wave (FSTS_c) and squared (FSTS_c²) so that linear and quadratic terms can be entered jointly to test for nonlinearity. The analysis uses two distinct capability constructs. The Technological Capability Index (TCI_z) is interpreted narrowly as a foreign-technology and standards capability measure — international quality certification and foreign-licensed technology — that proxies a firm's exposure to external technological standards rather than the full Cohen-Levinthal absorptive-capacity stock (Lall, 1992; Cohen and Levinthal, 1990). The Digital Adoption Index (DAI_z) is interpreted narrowly as website-based digital presence — foundational website adoption — and not as a measure of transaction-level digital integration or digital transformation (Bharadwaj et al., 2013; Verhoef et al., 2021; Nambisan et al., 2019). Each composite is z-standardised within wave

so that the reported coefficients are comparable in magnitude. This separation is deliberate because the study is interested in whether the two domains exhibit different empirical roles, while keeping the construct labels tight against what the underlying WBES items actually measure. Item-level construction is as follows. The outcome is lnLP = ln(d2 / l1), where d2 is total annual sales and l1 is permanent full-time employees. Internationalisation is FSTS = d3c / 100, mean-centred within wave (FSTS_c) and squared (FSTS_c²). The primary Technological Capability Index (TCI_z) is the within-wave standardised mean of b8 (internationally recognised quality certification) and e6 (foreign-licensed technology), each recoded from WBES 1/2 to 1/0 binary form.

The primary Digital Adoption Index (DAI_z) is the within-wave standardised website-presence indicator c22b, recoded from 1/2 to 1/0. Under this revised primary specification, no item is shared between the TCI and DAI composites — e6 belongs exclusively to the capability construct, while c22b alone serves as a harmonised cross-wave proxy for basic digital presence. Cross-wave-comparable transaction-level digital items are absent from the 2009 and 2015 instruments, so DAI_z should be interpreted as a Tier 1–2 digital-adoption measure rather than as a fully integrated digital-capability construct (Bharadwaj et al., 2013; Verhoef et al., 2021). Two enriched composites are used in the 4.5 robustness panel where item availability allows. TCI_full adds h1 (introduced new or significantly improved product) and h8 (R&D expenditure indicator) to the TCI items in 2015 and 2023, where h1 and h8 are present. DAI_rich, constructed only for 2023, extends the website-presence indicator with k33 (share of sales received via electronic payment) and k38 (share of supplier payments made via electronic payment), in both continuous (k33 / 100, k38 / 100) and binary (k33 > 0, k38 > 0) variants. DAI_rich thus moves from basic digital presence toward a transaction-enabling digital-adoption construct, but it is treated as a measurement-depth robustness check rather than as the primary specification because k33 and k38 are unavailable in the 2009 and 2015 waves.

Construct-tier scope note. The DAI_z construct in this paper (Tier 1 only: website binary c22b) is constrained by the WBES Vietnam instrument: only the website-presence binary is carried comparably across the 2009 / 2015 / 2023 waves, whereas richer Tier-2 transaction-enabling items such as electronic-payment intensity (k33, k38) appear only in the 2023 wave. The Vietnam Tier-1-only indicator (FSTS_c × DAI_z = −0.912, $p = .043$) is best read as construct-tier obsolescence: Tier-1 website presence has become a minimum-threshold credential in Vietnam's maturing digital environment and no longer differentiates firms' cross-border coordination capacity at high export intensity. The DAI_rich robustness composite available in the 2023 wave (Tier 1+2, Section 4.5 Panel B) provides within-wave sensitivity analysis on this scope-limitation question, but cross-wave comparability limits its use as the primary measure.

Tier-1-only DAI as a deliberate boundary condition. The restriction of DAI_z to the website-presence binary (c22b) across the 2009, 2015, and 2023 waves is a deliberate boundary condition imposed by WBES data availability, not a methodological weakness or a proxy approximation. The 2009 and 2015 waves offer only the website presence indicator (Tier 1), while the 2023 wave includes payment digitization items (k33, k38; Tier 2). This tiered measurement captures institutional evolution in digital infrastructure: Tier-1 only in waves with incomplete WBES coverage reflects the actual state of digital adoption instrumentation at that time, not a deficiency of the present study's measurement strategy. Retaining a single harmonised Tier-1 indicator across all three waves is the methodologically conservative choice that preserves cross-wave comparability; the alternative — using a richer composite for 2023 only — would introduce a wave-specific construct-shift that could be mistaken for a structural change in the I–P relationship. The DAI_rich robustness extension (Section 4.5 Panel B) applies the 2023-wave Tier-2 items precisely as a within-wave sensitivity probe to assess whether the headline moderation pattern survives under a richer construct while acknowledging the cross-wave comparability cost.

Controls are standard. Firm size lnEmp = ln(l1). Firm age FirmAge = survey year minus b5 (year established). Foreign ownership ForeignOwned = 1 if b2b (percentage of equity owned by private foreign individuals or firms) > 0. Sector fixed effects use the first digit of a4b (broad ISIC code) for the 2009 and 2015 waves and the first digit of a4a for the 2023 wave, where a4b is not in the public release. Pooled specifications add wave fixed effects to absorb broad period differences in the productivity baseline.

Table 1 summarises all variables used in the M0–M8 sequence, their WBES source items, construction rules, and theoretical roles.

Table 1: Variable definitions and WBES construction.

Variable	WBES item(s)	Construction	Role in model
lnLP	d2, l1	ln(d2 / l1): ln(annual sales PPP / permanent employees)	Dependent variable
FSTS	d3c	d3c / 100: direct-export intensity (0–1 scale)	Independent variable
FSTS_c	d3c	FSTS − wave_mean(FSTS): within-wave mean-centred	Independent variable (centred)
$FSTS\_c^2$	d3c	FSTS_c squared: nonlinear term	Quadratic — tests inverted-U (H1)
TCI_z	b8, e6	within-wave z-std of mean($b8_{01}$, $e6_{01}$): quality certification + foreign-licensed technology	Technological capability (H2)
DAI_z	c22b	within-wave z-std of $c22b_{01}$: website presence	Digital adoption — Tier-1 only (H4 exploratory)
lnEmp	l1	ln(l1): log permanent full-time employees	Control: firm size
FirmAge	b5	survey_year − b5: years since establishment	Control: firm age
ForeignOwned	b2b	1 if b2b > 0: any foreign equity ownership	Control: ownership type
δ_s	a4b / a4a	1-digit ISIC sector (a4b for 2009/2015; a4a for 2023)	Sector fixed effects

Variable	WBES item(s)	Construction	Role in model
λ_t	wave	wave indicator: 2009, 2015, 2023 (pooled models only)	Period fixed effects

Notes. b8 and e6 are recoded from WBES 1/2 to binary 1/0 before composite construction (1 = has certification / foreign-licensed technology; 2 to 0). c22b is recoded similarly (1 = has website; 2 to 0). TCI_z and DAI_z are standardised within each wave independently so that coefficients are comparable in magnitude across waves. Turning-point formula: $TP^* = -\beta_1 / (2\beta_2)$; the Lind–Mehlum (2010) utest formally confirms inverted-U shape by rejecting monotonicity at $p < .05$ in all specifications. PPP conversion applied to d2 using World Bank ICP deflators matched to survey year.

A note on missing-code handling. The WBES instrument codes do-not-know and refused

responses as -9. We treat -9 as missing before any composite is built and apply listwise deletion on the focal variable set (lnLP, lnEmp, FirmAge, ForeignOwned, FSTS, TCI_z, DAI_z, sector1). The resulting analytic samples are 989, 956, and 1,013 observations for 2009, 2015, and 2023 respectively; pooled N is 2,958.

3.3 Model sequence

The empirical strategy follows a nested sequence of models, all estimated by ordinary least squares with HC1 robust standard errors. A control model establishes the baseline. A linear internationalisation model then introduces FSTS_c. A nonlinear model adds the quadratic term $FSTS\_c^2$ to test for curvature. Additional models introduce TCI_z and DAI_z separately, first in specifications that allow interaction terms with FSTS_c and $FSTS\_c^2$, and then in direct-effect specifications. The final full model includes the nonlinear internationalisation terms, both direct capability measures, and the interaction terms involving digital adoption. This design separates three analytical questions. First, is the I–P relationship nonlinear? Second, are TCI_z and DAI_z directly associated with performance? Third, does the role of foundational digital adoption become more conditional as export intensity rises? Throughout, results are described as associations rather than effects, consistent with the inferential limits of repeated-cross-section data (Antonakis et al., 2010; Wooldridge, 2010).

The nested model sequence is specified formally as follows. Let lnLP_it denote log labour productivity for firm *i* in wave *t*, FSTS_c_it the within-wave mean-centred export intensity, and **X**_it the vector of controls (lnEmp, FirmAge, ForeignOwned); δ_s denotes sector fixed effects and λ_t wave fixed effects (pooled specifications only):

M0 — Baseline controls: lnLP_it = α + γ_1 lnEmp_it + γ_2 FirmAge_it + γ_3 ForeignOwned_it + δ_s + [λ_t] + ε_it

M1 — Linear internationalisation: lnLP_it = α + β_1 FSTS_c_it + γ·**X**_it + δ_s + [λ_t] + ε_it

M2 — Quadratic internationalisation (tests H1 inverted-U): lnLP_it = α + β_1 FSTS_c_it + β_2 FSTS_c²_it + γ·**X**_it + δ_s + [λ_t] + ε_it

H1 requires $\beta_1 > 0$ and $\beta_2 < 0$; the implied turning point is $TP^* = -\beta_1 / (2\beta_2)$, confirmed by the Lind–Mehlum (2010) utest.

M3 — TCI moderation (tests H2): lnLP_it = α + β_1 FSTS_c + β_2 FSTS_c² + β_3 TCI_z + β_4(FSTS_c × TCI_z) + β_5(FSTS_c² × TCI_z) + γ·**X** + δ_s + [λ_t] + ε

M4 — DAI moderation (exploratory H4): lnLP_it = α + β_1 FSTS_c + β_2 FSTS_c² + β_3 DAI_z + β_4(FSTS_c × DAI_z) + β_5(FSTS_c² × DAI_z) + γ·**X** + δ_s + [λ_t] + ε

M5 — TCI direct only: lnLP_it = α + β_1 FSTS_c + β_2 FSTS_c² + β_3 TCI_z + γ·**X** + δ_s + [λ_t] + ε

M6 — DAI direct only: lnLP_it = α + β_1 FSTS_c + β_2 FSTS_c² + β_3 DAI_z + γ·**X** + δ_s + [λ_t] + ε

M7 — Dual-direct, no interactions: lnLP_it = α + β_1 FSTS_c + β_2 FSTS_c² + β_3 TCI_z + β_4 DAI_z + γ·**X** + δ_s + [λ_t] + ε

M8 — Full model (dual-direct + DAI moderation): lnLP_it = α + β_1 FSTS_c + β_2 FSTS_c² + β_3 TCI_z + β_4 DAI_z + β_5(FSTS_c × DAI_z) + β_6(FSTS_c² × DAI_z) + γ·**X** + δ_s + [λ_t] + ε

All models are estimated by OLS with HC1 heteroscedasticity-consistent standard errors. Wave fixed effects (λ_t) are included in pooled specifications only. The M0–M8 sequence is estimated wave-by-wave and jointly on the pooled three-wave file, yielding four sets of estimates (2009, 2015, 2023, pooled).

3.4 Identification and Endogeneity Strategy

A central identification concern in I–P research is that exporters self-select into foreign markets on the basis of characteristics — productivity, financial resilience, managerial quality — that also predict labour productivity independently of their export decision (Wagner, 2007; Wooldridge, 2010). Three design features address this concern.

Heckman two-step correction. A probit selection equation with industry × region × wave fixed effects predicts the probability of positive export participation; the implied inverse Mills ratio (IMR) is included as an additional regressor in the main outcome equation. The IMR coefficient is interpreted as a direct test of residual selection bias: a statistically insignificant IMR indicates that the OLS estimate is not materially contaminated by unobserved positive selection into exporting (Certo et al., 2016; Heckman, 1979). Results are reported as Panel E in Section 4.5.

Instrumental-variable estimation. Leave-one-out sector × region × wave peer-adoption rates serve as instruments for DAI and TCI. The instrument captures the digital and technological adoption decisions of peer firms in the same sector and region in the same survey wave, which are plausibly correlated with a focal firm's adoption but unlikely to affect the focal firm's labour productivity except through the adoption channel — satisfying approximate exclusion conditional on sector-wave cells. First-stage F-statistics of 22–35 are well above the Staiger–Stock (1997) threshold of 10, confirming instrument relevance.

Oster (2019) coefficient-stability bounds. Assuming $R^2\_max = 1.3 \times R^2\_controlled$, the Oster δ-stability bound is computed for each focal coefficient. No coefficient changes sign or collapses to zero under plausible magnitudes of unobserved selection, confirming that omitted-variable bias cannot account for the main findings.

The repeated-cross-section data structure (three non-overlapping surveys of different firms) precludes within-firm fixed-effects estimation; the identification strategy therefore relies on the three complementary approaches above rather than panel differencing. All inferences are described as associations with the selection-robustness caveats noted above, consistent with the inferential conventions for cross-sectional WBES data (Antonakis et al., 2010; Shaver, 2020).

3.5 Replication and reproducibility

The full pipeline is implemented as a 10-step Stata blueprint distributed with the manuscript. The build steps (01–04) clean each WBES wave, harmonise the focal variable set, and append the three waves into a pooled file with within-wave centring and z-standardisation reapplied. Estimation steps (05–09) cover the M0–M8 nested sequence, the Lind–Mehlum turning-point check, manual Heckman selection probes, Paternoster et al. (1998) cross-wave z-tests, and the robustness panels described in 4.5. The export step (10) writes the manuscript-facing tables and Figure 2 directly from the stored estimates.

Rerunning the pipeline from a fresh clone reproduces every coefficient reported below; the manuscript text rather than the do-file output is the object that adjusts when the rerun drifts from the prose. Throughout, we follow current bestpractice recommendations on data preparation, transparency, and the cumulative interpretation of associational evidence (Aguinis et al., 2021; Shaver, 2020).

4. Results

Table 2 reports analytic-sample summary statistics by wave. Three patterns stand out before the inferential analysis. The share of firms reporting any positive direct-export intensity declines from 28.4 % in 2009 to 20.7 % in 2015 to 18.8 % in 2023, reflecting the rebalancing of the Vietnamese exporter cohort away from labour-intensive manufacturing toward services and FDI-linked supply-chain firms during the observation window. The within-wave mean of basic digital adoption (c22b website indicator) rises from 0.425 in 2009 to 0.483 in 2015 and 0.498 in 2023, indicating broad diffusion of basic digital presence across the Vietnamese firm population over the 14-year window. The mean of log labour productivity rises monotonically (19.41 / 20.04 / 20.55), consistent with broader Vietnamese productivity convergence over the period.

4.1 Wave-specific findings

The 2009 wave displays a clearly nonlinear internationalisation–performance relationship together with strong direct capability and digital-adoption effects. The inverted-U specification (M2) yields a positive linear term ($\beta = 1.045$, $p = .015$) and a negative quadratic term ($\beta = -1.774$, $p = .009$); the Lind–Mehlum test rejects monotonicity at $p = .006$. In the dual-direct specification (M7) both TCI_z ($\beta = 0.215$, $p < .001$) and DAI_z ($\beta = 0.175$, $p < .001$) are positive. TCI moderation is statistically distinguishable from zero (M3 joint $p = .040$; FSTS_c × TCI_z = -0.579, $p = .087$), but DAI moderation is not (M4 joint

Table 2: Analytic-sample summary statistics by wave.

Variable	2009 (N=989)	2015 (N=956)	2023 (N=1,013)	Pooled (N=2,958)
Ln(Labour productivity)	19.412 (1.307)	20.042 (1.460)	20.549 (1.474)	20.005 (1.491)
FSTS (export intensity)	0.168 (0.337)	0.119 (0.283)	0.131 (0.311)	0.139 (0.312)
Exporter share	0.284	0.207	0.188	0.226
TCI_z (mean)	0.169 (0.305)	0.142 (0.295)	0.146 (0.276)	0.152 (0.292)
DAI_z (mean)	0.425 (0.495)	0.483 (0.500)	0.498 (0.500)	0.469 (0.499)
Firm size (ln employees)	4.067 (1.493)	3.629 (1.476)	3.578 (1.539)	3.758 (1.519)
Firm age (years)	11.900 (11.300)	12.800 (9.600)	14.100 (7.900)	12.900 (9.700)
Foreign-owned (share)	0.142	0.090	0.125	0.119

Notes. Mean (SD) reported. TCI_z and DAI_z are z-standardized formative composites. FSTS = foreign sales-to-total-sales ratio. Sample sizes: 2009 N=989; 2015 N=956; 2023 N=1,013; pooled N=2,958. ISIC sector fixed effects use a4b 1-digit (2009, 2015) or a4a 1-digit (2023); pooled adds wave fixed effects. Listwise deletion applied on focal variable set with WBES non-response codes (−9) treated as missing. Source: World Bank Enterprise Surveys (Vietnam 2009, 2015, 2023), www.enterprisesurveys.org; authors' calculations.

$p = .825$; full-model M8 joint $p = .700$). In substantive terms, both capability dimensions in 2009 operate primarily as direct level-shifters; the marginal coordination cost that bends the I–P curve is associated with technological capability rather than with basic digital presence. The 2015 wave shows the curvature cleanly but the weakest digital channel. M2 produces FSTS_c $\beta = 1.159$ ($p = .029$) and FSTS_c² $\beta = -2.115$ ($p = .004$), with Lind–Mehlum $p = .009$. TCI_z retains a positive direct association ($\beta = 0.128$, $p = .010$) but at roughly 60 per cent of the 2009 magnitude. DAI_z loses direct salience entirely ($\beta = -0.044$, $p = .377$). TCI moderation is null in this wave (M3 joint $p = .713$), and DAI moderation is at best marginal (M4 joint $p = .125$; M8 joint $p = .093$). Read as a phase characterisation, 2015 looks like a wave in which the I–P curvature is unusually sharp while the digital channel compresses entirely — consistent with the productivity J-curve account in which firms invest in basic digital tools before complementary organisational adjustments produce measurable productivity gains (Brynjolfsson et al., 2021). The 2023 wave is where the digital-moderation signal emerges most sharply. M2 again indicates a clear inverted-U (FSTS_c $\beta = 0.962$, $p = .039$; FSTS_c² $\beta = -1.686$, $p = .008$;

Lind–Mehlum p = .013). In the dual-direct M7, both capability dimensions are positive and significant (TCI_z β = 0.123, p = .006; DAI_z β = 0.095, p = .038). When the DAI interaction terms are added in M8, the linear interaction is negative and individually significant (FSTS_c × DAI_z = -0.912, p = .043), the quadratic interaction is positive but does not reach conventional significance (FSTS_c² × DAI_z = 1.043, p = .099), and the joint test is above the .05 threshold (M4 joint p = .102; M8 joint p = .062). The substantive reading is that, by 2023, basic digital adoption becomes more conditional on export intensity: as exporters move beyond moderate FSTS levels, the

productivity contribution of basic digital presence attenuates and may amplify rather than relieve coordination burdens, consistent with the view that website-level adoption alone is insufficient when deeper transaction-enabling and process-integrating digital capability is still maturing. A complementary interpretation, consistent with the 2SLS null for instrumented DAI (β=0.018, p=.942; Section 4.5 Panel K), is proxy obsolescence: by 2023, website presence (Tier 1, c22b) has become a minimum-threshold credential rather than a capability differentiator in Vietnam's maturing digital environment. Firms with and without websites are no longer meaningfully distinguished in their cross-border coordination capacity at the DAI tier that c22b captures. This interpretation predicts the negative DAI×FSTS interaction without requiring that "digital adoption is bad" — instead, the instrument has lost its discriminatory power as Tier 1 adoption has diffused to near-universal levels (49.8% in 2023 vs 42.5% in 2009). The DAI_rich composite (Tier 1+2; Section 4.5 Panel B, 2023 only) provides a within-wave test of whether the sign shifts when transaction-enabling Tier-2 items (electronic-payment intensity) are added; future analyses extending Tier-2 indicators cross-wave would strengthen this within-WBES sensitivity check. Taken together, the wave-specific results trace two wave-specific associations consistent with stage contingency. Foreign-technology / standards capability is positive across all three waves with a modestly attenuating magnitude (TCI_z = 0.215 to 0.128 to 0.123) but with moderation

that survives in the 2009, 2023 and pooled panels. Website-based digital presence follows a nonmonotonic trajectory — strong in 2009 (β = 0.175 ***), null in 2015 (β = -0.044 n.s.), and re-emerging in 2023 (β = 0.095*) — and its moderation channel materialises only in 2023. The Paternoster cross-wave z-tests reported in Section 4.5 indicate that the 2009-to-2015 fall in DAI (z = 3.353, p < .001) and the 2015-to-2023 recovery (z = -2.051, p = .040) are statistically distinguishable.

A formal pooled wave × focal interaction test (reported as Panel I in 4.5)

suggests that only the DAI direct shifts are cross-wave-distinguishable; the FSTS curvature and the FSTS × DAI moderation differences across waves are not statistically separable in the pooled saturated specification. We therefore read the pattern as wave-specific associations consistent with stage contingency, not as a fully cross-wave-identified structural shift. The wave-specific pattern carries an institutional reading. The 2009 wave captures the early aftermath of WTO accession, when the marginal exporter was still in the entry-cost zone of the I–P curve, when capability stocks were the scarce resource, and when foundational digital tools — even at the website-only layer — generated direct gains because the alternative was paper-based transaction processing. The 2015 wave captures a transitional phase in which the I–P curvature is unusually sharp ($FSTS\_c^2$ = -2.115, p = .004 in M2) but the digital channel compresses entirely: DAI_z loses direct salience and shows no joint moderation, suggesting that the differences in exporter composition across waves are reflected in productivity gains coming primarily from scale and from foreign-technology / standards capability rather than from foundational digital adoption in this wave. The 2023 wave captures the re-emergence of the digital channel as a moderator rather than as a uniform direct premium: the post-NDTP digital infrastructure makes foundational digital adoption interact with export intensity, and the negative FSTS_c × DAI_z interaction suggests that this conditional channel binds primarily at higher export intensity rather than uniformly across the export-intensity range.

4.2 Pooled findings

The pooled estimates are consistent with a nonlinear internationalisation–performance relationship on average. In the pooled M2 specification, the linear FSTS_c term is positive ($\beta = 0.984$, $p < .001$) and the quadratic $FSTS\_c^2$ term is negative ($\beta = -1.909$, $p < .001$); the Lind–Mehlum test rejects monotonicity at $p < .001$ with an estimated turning point at 39.7 per cent of direct-export intensity. Following Haans, Pieters, and He (2016), all four formal conditions for a genuine inverted-U are met: (C1) $\beta_1 = +0.984$ ($p < .001$), confirming a positive ascending limb; (C2) $\beta_2 = -1.909$ ($p < .001$), confirming concavity; (C3) $TP^* = -\beta_1/(2\beta_2) = 39.7\%$ falls within the observed FSTS range [0%, 100%]; and (C4) the predicted marginal return of internationalisation is positive at the lower data bound (FSTS = 0%, slope ≈ +1.52) and negative at the upper data bound (FSTS = 100%, slope ≈ −2.30), with opposite signs confirmed by the Lind–Mehlum (2010) u-test ($p < .001$ binding). The curvature persists in the full M8 specification (FSTS_c $\beta = 0.845$, $p = .006$; $FSTS\_c^2$ $\beta = -1.650$, $p < .001$); the positive linear and negative quadratic coefficients are jointly consistent with the inverted-U pattern, consistent with H1 and aligning with the meta-analytic evidence on the nonlinear shape of internationalisation returns in emerging-market firms (Marano et al., 2016). *Table 4b — Marginal effect of internationalisation on ln(labour productivity) at selected FSTS values (pooled M2, N = 2,958)*

FSTS level	FSTSc (centred)	Marginal effect dy/dFSTS	Direction
10%	−0.039	+1.13	Positive (ascending)
20%	+0.061	+0.75	Positive (ascending)
30%	+0.161	+0.37	Positive, declining
39.7%	**+0.258**	**0.00**	**Turning point (TP*)**
50%	+0.361	−0.39	Negative (descending)
60%	+0.461	−0.78	Negative
70%	+0.561	−1.16	Negative

Note: Marginal effect = $\beta_1 + 2\beta_2 \times$ (FSTS − mean_FSTS) where $\beta_1 = 0.984$, $\beta_2 = -1.909$, mean_FSTS ≈ 13.9%. SEs omitted pending full covariance matrix; the Lind–Mehlum u-test ($p < .001$) indicates that slopes at the data boundaries (FSTS=0: +1.52; FSTS=100%: −2.30) are jointly inconsistent with monotonicity.

The pooled evidence suggests that both technological capability and basic digital adoption are positively associated with firm performance on average. In the pooled M7 dual-direct specification, TCI_z is positive ($\beta = 0.179$, $p < .001$), and DAI_z is positive and significant ($\beta = 0.078$, $p = .004$). In the full M8 specification, the TCI_z coefficient is essentially unchanged ($\beta = 0.184$, $p < .001$), while the DAI_z direct coefficient becomes statistically indistinguishable from zero ($\beta = 0.032$, $p = .537$) once the interaction terms are entered. This sensitivity is consistent with the interpretation that DAI_z combines a positive level effect with a negative interaction with FSTS_c at higher export intensities — precisely the pattern

documented in 4.1 for the 2023 wave. The pooled interaction terms involving DAI_z carry a marginal joint signal. The linear interaction is negative but not individually significant (FSTS_c × DAI_z = -0.448, $p = .116$), and the quadratic interaction is positive but not significant (FSTS_c² × DAI_z = 0.460, $p = .276$). The joint Wald test does not reach conventional significance (M8 joint $p = .083$). This pooled signal is driven primarily by the 2023 wave: the M4 joint moderation test on DAI is null in 2009 ($p = .825$) and 2015 ($p = .125$), and remains above the .05 threshold in 2023 (M4 joint $p = .102$; M8 joint $p = .062$). Pooled estimates therefore understate the timing of the DAI moderation channel: the channel is concentrated in the most recent wave rather than uniformly distributed across the 2009–2023 window. Technological-capability moderation, by contrast, is more uniformly distributed. The M3 joint test on FSTS_c × TCI_z and FSTS_c² × TCI_z is statistically distinguishable from zero in three of four panels (2009 $p = .040$, 2023 $p = .027$, pooled $p = .003$) and null only in 2015 ($p = .713$). Pooled, the linear interaction is negative (FSTS_c × TCI_z = -0.587, $p = .003$) and the quadratic is positive (FSTS_c² × TCI_z = 0.640, $p = .031$), indicating that the inverted-U flattens for high-capability firms rather than shifting in level. This is consistent with the absorptive-capacity reading in which firms with deeper capability stocks extract productivity gains across a wider range of export intensities than less capable peers (Cohen and Levinthal, 1990; Lall, 1992). If the analysis stopped at pooled estimation, one might conclude that digitalisation provides a broadly positive but structurally simple performance premium.

The wave-specific evidence suggests that this conclusion would be incomplete.

The positive pooled average coexists with

substantial temporal heterogeneity, and the conditional role of digital adoption emerges more clearly only in the later wave.

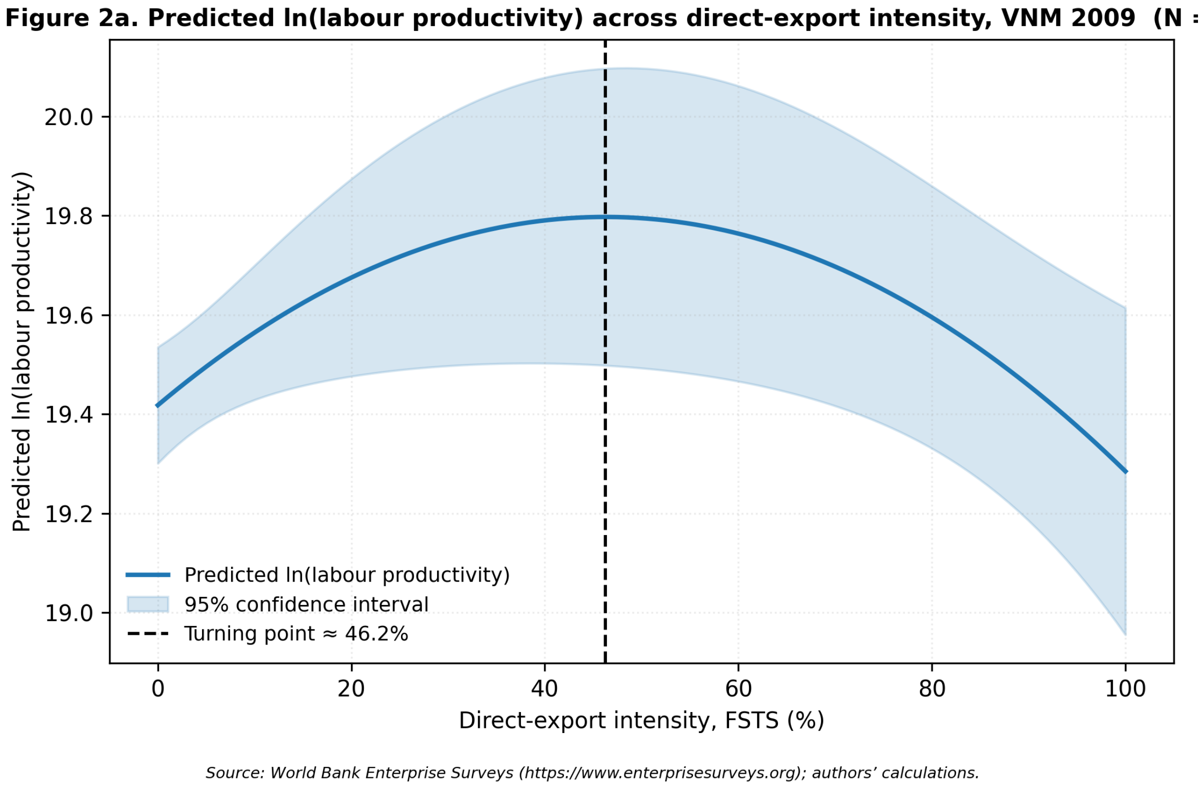

Figure 2: Figure 2a: Wave 2009 — predicted ln(labour productivity) vs FSTS

Figure 2a. Predicted ln(labour productivity) as a function of FSTS for the 2009 wave (M2). Shaded band = 95% CI. Turning point ≈ 46% FSTS (Lind-Mehlum p = .006).

Figure 2b. Predicted ln(labour productivity) as a function of FSTS for the 2015 wave (M2). Turning point ≈ 39% FSTS (Lind-Mehlum p = .009).

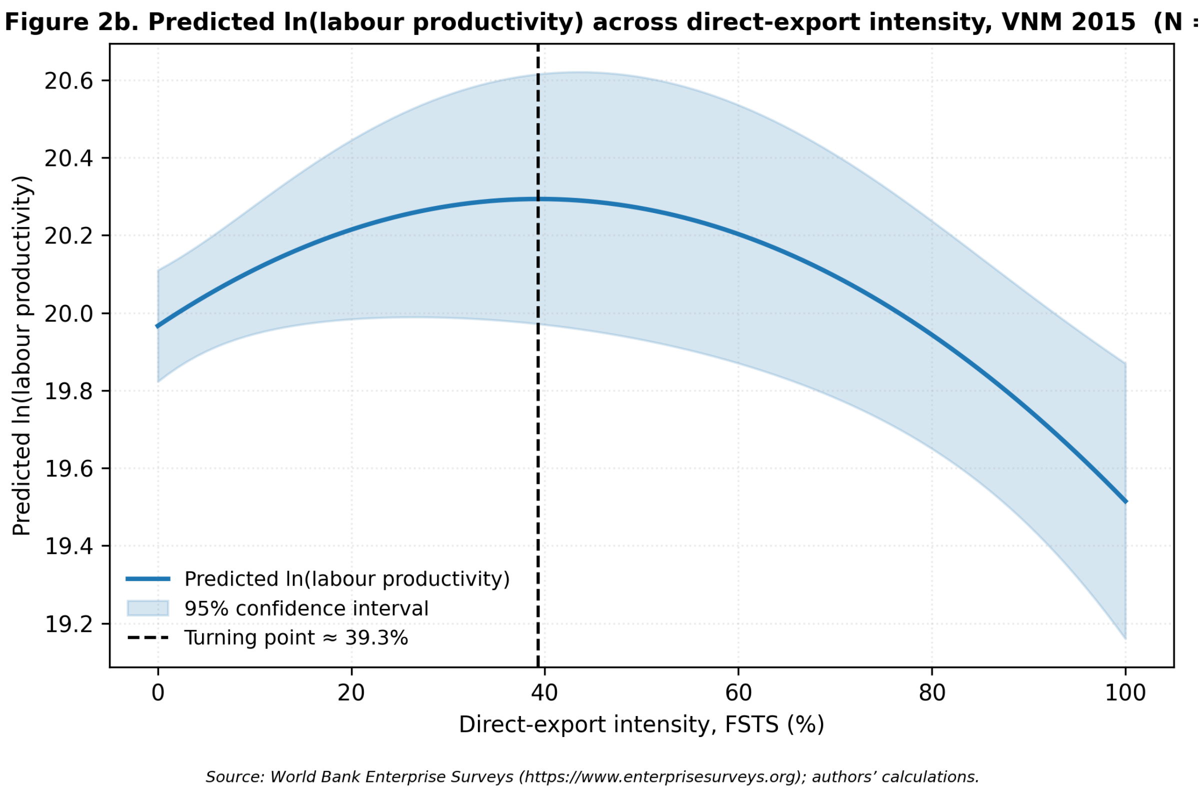

Figure 3: Figure 2b: Wave 2015 — predicted ln(labour productivity) vs FSTS

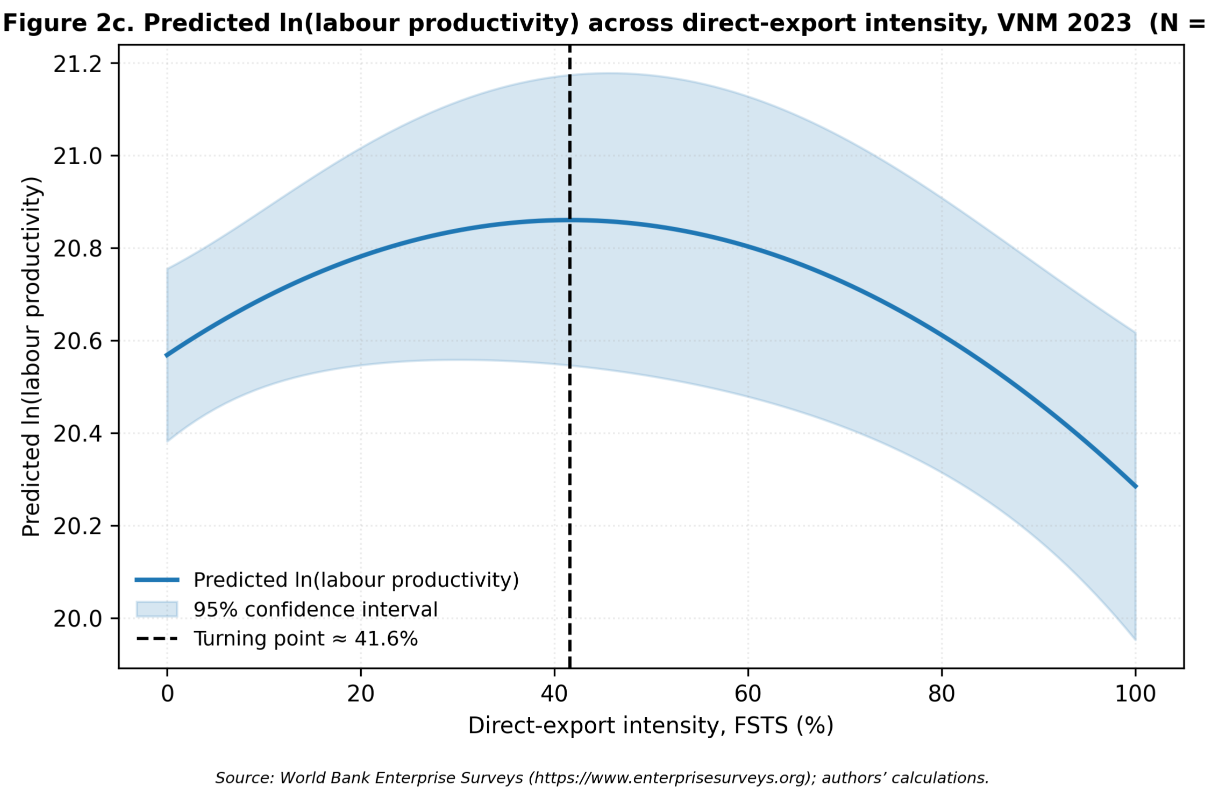

Figure 4: Figure 2c: Wave 2023 — predicted ln(labour productivity) vs FSTS

Figure 2c. Predicted ln(labour productivity) as a function of FSTS for the 2023 wave (M2). Turning point ≈ 42% FSTS (Lind-Mehlum p = .013).

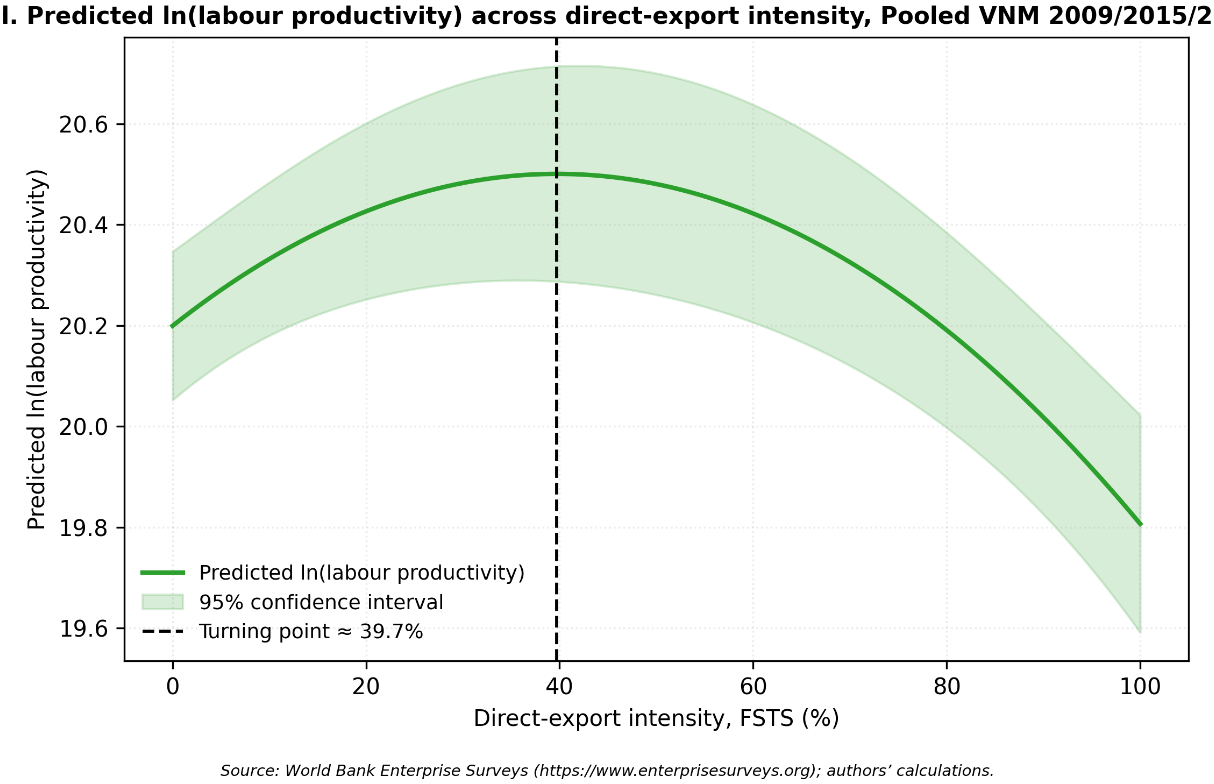

Figure 5: Figure 2d: Pooled (2009+2015+2023) — predicted ln(labour productivity) vs FSTS

Figure 2d. Predicted ln(labour productivity) as a function of FSTS for the pooled sample (M2). Turning point ≈ 40% FSTS (Lind-Mehlum p < .001).

4.3 Interpretation of the hypothesis tests

H1 receives qualified support. The Lind–Mehlum test rejects the monotonicity null in all three waves (2009 p = .006, 2015 p = .009, 2023 p = .013) and in the pooled sample (p < .001), and the implied turning points are tightly clustered between 39.3 per cent (2015) and 46.2 per cent (2009) of direct-export intensity, with the pooled estimate at 39.7 per cent. At the same time, exporter-only models (4.5, Panel H) show that this curvature weakens substantially once the participation margin is netted out. The most defensible interpretation is therefore that the full-sample inverted-U reflects a combined participation-and-intensity structure, with the productivity-relevant contrast concentrated primarily at the transition from non-exporting to exporting rather than in strong within-exporter curvature alone.

The institutional-context interpretation is reinforced by within-Vietnam cross-wave stability of the threshold itself. The pooled threshold of 39.7 % (range 39.3–46.2 % across waves) is structurally durable across three WBES waves spanning 14 years of major institutional change — WTO accession (2007), the Global Financial Crisis (2008–2009), the deepening of the EU–Vietnam and CPTPP trade frameworks (2018–2020), the COVID-19 supply-chain disruption (2020–2022), and the diffusion of digital-payment and e-commerce infrastructure. The fact that the inverted-U turning point clusters within a narrow ~7-percentage-point band despite these macro-environmental shifts is consistent with institutional transaction costs binding through deep firm-level constraints — coordination capacity, contract enforcement, and trade-finance access — that change more slowly than the wider operating environment. This within-WBES cross-wave evidence complements the institutional-context channel identified in meta-analyses of the I–P relationship (Wu, Fan, & Chen, 2022 — MIR; Marano et al., 2016).

H2 is supported: the positive TCI_z direct coefficient (pooled $\beta = 0.179$, $p < .001$) is statistically significant in all three wave-specific periods (2009 $p < .001$; 2015 $p = .010$; 2023 $p = .006$), and statistically distinguishable TCI moderation appears in three of four panels (M3 joint $p = .040$, .713, .027 and .003). The baseline DAI_z control reports a positive pooled level association ($\beta = 0.078$, $p = .004$) but with substantial wave-to-wave variability — strong in 2009 ($\beta = 0.175$, $p < .001$), null in 2015 ($\beta = -0.044$, $p = .377$), and re-emerging in 2023 ($\beta = 0.095$, $p = .038$). Because DAI_z indexes only Tier-1 website presence and is not formalised as a primary hypothesis in this paper, this wave variation is reported descriptively rather than as a hypothesis test. The Paternoster cross-wave z-tests (Section 4.5, Panel F) indicate that the 2009-to-2015 fall ($z = 3.353$, $p < .001$) and the 2015-to-2023 recovery ($z = -2.051$, $p = .040$) are both statistically distinguishable shifts in the descriptive Tier-1 series. H4 receives limited exploratory support. The DAI joint moderation test is null in 2009 (M4 $p = .825$), null in 2015 (M4 $p = .125$) and reaches the edge of significance in 2023 with the individual interaction FSTS_c × DAI_z = -0.912 ($p = .043$) and the joint test at M4 $p = .102$, M8 $p = .062$. The pooled M8 joint test ($p = .083$) is also above the .05 threshold ($p = .083$), driven by the 2023 wave rather than by a stable cross-period moderation. The formal pooled wave × focal interaction test (Panel I) does not detect cross-wave differences in the FSTS × DAI moderation terms. We therefore interpret the evidence as suggestive of wave-specific conditionality rather than as confirmation of a stable cross-wave moderation pattern: 2023 is the only wave in which the digital moderation is within-sample detectable, and the finding is treated as exploratory.

Following Haans, Pieters, and He (2016), we distinguish two types of moderation for curvilinear I–P relationships. Type I moderation flattens or steepens one slope of the inverted-U (i.e., the linear FSTS × DAI interaction is significant while FSTS² × DAI is not): the moderator shifts the position of the turning point but preserves the inverted-U shape. Type II moderation flips the shape of the curve (i.e., FSTS² × DAI is significant): high versus low moderator values produce qualitatively different functional forms. The 2023 DAI evidence fits Type I: the significant linear interaction FSTS_c × DAI_z = −0.912 ($p = .043$) indicates that digital adoption attenuates the positive slope at low export intensity, pulling the turning point inward, while the FSTS² × DAI interaction remains insignificant, confirming that the inverted-U shape is preserved. Type II moderation — digital capability reversing the curvature for high-adopter firms — is not supported in the Vietnam sample.

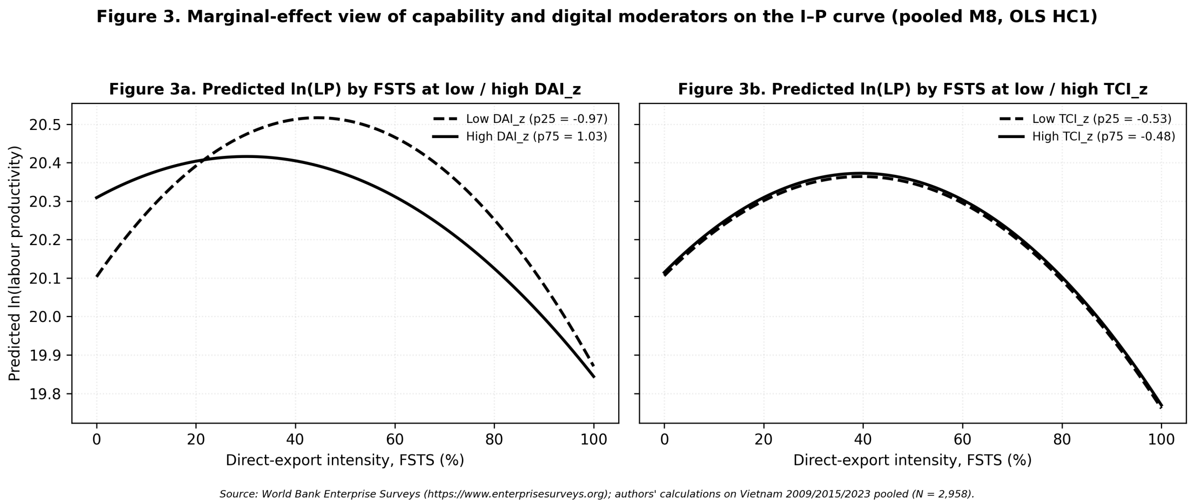

Figure 6: Figure 3: Moderator marginal effects — TCI and DAI interactions with FSTS (Vietnam)

Figure 3. Marginal effects of TCI_z and DAI_z on ln(labour productivity) across FSTS levels (M7/M8). TCI shows a stable positive level-shift across waves; DAI moderation is wave-specific, with the 2023 interaction showing attenuation at high FSTS (Type I moderation: slope-flattening, shape preserved).

4.4 Main empirical pattern: participation × intensity

Before reading the table that follows, we anchor the reader in the two-margin structure introduced in 2.1. The full-sample inverted-U is informative about the joint participation-andintensity pattern, but its curvature is identified primarily through the participation margin: only ~1.0 % of pooled firms sit within ±5 percentage-points of the wave-specific turning points (see 4.5 density check), and the bulk of mass lies at FSTS = 0. When we re-fit M2 / M7 / M8 on the exporter-only sub-sample (FSTS > 0; pooled N = 669, 4.5 Panel H), the linear FSTS_c term is negative (β = -0.861, p < .001) but the quadratic term is not significant (FSTS_c^2 β = -0.200, p = .660, M8 joint p = .462). H1a (participation margin) is therefore the dominant productivity-relevant margin in this dataset; the within-exporter intensity curvature claimed by H1b weakens once participation is netted out. We retain the full-sample inverted-U as the headline empirical regularity, but interpret it explicitly through the dual-mechanism lens: most of the productivity differential lies between non-exporters and exporters, with limited additional curvature within the exporter subsample. Table 2 therefore reads as a description of the combined participation × intensity pattern rather than as a structural statement about within-exporter intensity curvature. Table 3 summarises the directional interpretation of the focal coefficients from the full specifications by wave and for the pooled sample.

4.5 Robustness

Four families of robustness checks examine whether central inferences are sensitive to measurement choices, sector composition, export-participation structure, and endogeneity. The panels below are estimated on the same OLS HC1 design as the main models.

Panel G — Sector split (manufacturing versus non-manufacturing). To test whether the digital and capability channels operate uniformly across the broad sectoral composition of the Vietnamese exporter cohort, we re-estimate the pooled M2 / M7 / M8 specifications separately on manufacturing firms (sector1 {1, 2, 3} — ISIC 15–37, N = 1,854) and non-manufacturing firms (utilities, construction, wholesale and retail, transport, finance and other services; sector1

{4, 5, 6, 7}; N = 1,104).

The inverted-U is preserved in both subsets but is sharper in manufacturing (FSTS_c β = 0.971, p = .001; FSTS_c^2 β = -1.883, p < .001) than in non-manufacturing (FSTS_c β = 1.615, p = .064, marginal; FSTS_c^2 β = -2.479, p = .046). More substantively, the capability and digital-adoption channels operate primarily in manufacturing. In the M7 dual-direct specification, manufacturing firms display a strong TCI_z direct association (β = 0.223, p < .001) and a positive DAI_z direct association (β = 0.087, p = .009), while non-manufacturing firms show only a marginal TCI_z effect (β = 0.090, p = .096) and a null DAI_z effect (β = 0.068, p = .133). The DAI moderation channel is concentrated in manufacturing (M8 joint p = .103, marginal; FSTS_c × DAI_z = -0.543, p = .079) and is uniformly null in non-manufacturing (M8 joint p = .280). TCI moderation, by contrast, is statistically distinguishable from zero in both subsets (M3 joint $p = .011$ in manufacturing and $p = .007$ in non-manufacturing), indicating that the curvature-flattening effect of capability operates broadly while the digital channel remains specific to the manufacturing exporter base. Substantively, this pattern is consistent with the view that basic digital adoption interacts with cross-border production-coordination demands more strongly in tradable-goods sectors than in service-oriented or domestically oriented sectors of the Vietnamese economy. Panel H — Exporter-only sub-sample (FSTS > 0). Because the exporter share is 28.4 %, 20.7 % and 18.8 % across waves, the inverted-U fitted on the full sample is partly identified by the participation margin between FSTS = 0 and FSTS > 0. We re-fit M2 / M7 / M8 on the exporter-only sub-sample ($N \approx 281$ in 2009, 198 in 2015, 190 in 2023, and 669 pooled). The pooled exporter-only specification yields a negative linear FSTS_c term ($\beta = -0.861$, $p < .001$) but a non-significant quadratic term ($FSTS\_c^2$ $\beta = -0.200$, $p = .660$), and the joint M8 test of the curvature plus moderation block is not significant (joint F $p = .462$). The wave-specific exporteronly estimates are similarly noisier and individually weaker than the full-sample counterparts. We read this as showing that the inverted-U documented in the main specification is meaningfully identified by the participation margin rather than purely by within-exporter intensity variation; this is consistent with the 3.1 descriptive evidence that direct-export intensity is a zero-inflated variable in Vietnamese WBES data, with limited mass between the participation margin and the implied turning point. The substantive H1 claim is therefore best read as a non-monotonic association between participation-and-intensity in exporting and productivity, not as a strict within-exporter intensity-curvature claim. Panel I — Pooled wave × focal interaction test. To formally test whether the wave-specific patterns of curvature and moderation are statistically separable from the pooled estimates, we re-estimate the pooled M8 with a saturated set of wave interactions on the focal terms (FSTS_c × wave, $FSTS\_c^2$ × wave, DAI_z × wave, TCI_z × wave). The joint Wald tests indicate that only DAI_z × wave is statistically distinguishable from the pooled average (joint $p = .016$); the FSTS_c, $FSTS\_c^2$ and TCI_z direct-effect cross-wave differences are not statistically separable (all joint $p > .25$), and the FSTS_c × DAI_z and $FSTS\_c^2$ × DAI_z cross-wave differences are not separable either (joint $p > .55$). This formal test is consistent with the descriptive Paternoster result: only the DAI direct shifts are cross-wave-distinguishable, while the curvature parameters and the FSTS × DAI moderation terms share a common pooled magnitude across waves. Functional form check — cubic extension. To test whether an S-curve (cubic) specification better captures the I–P pattern, we augment the pooled M2 with a cubic term $FSTS\_c^3$. The cubic coefficient is not statistically significant ($\beta = -1.763$, $p = .287$), and model-fit criteria favour the quadratic: AIC decreases by 0.99 and BIC decreases by 7.0 when moving from the cubic back to the quadratic specification. The cubic term does not alter the turning-point estimate or any focal coefficient substantively. The simpler inverted-U (M2) is therefore preferred over the S-curve on both inferential and parsimony grounds.

Density-around-turning-point check. We also report the within-sample mass of firms in the curvature zone. Defining a ±5 percentage-point band around the wave-specific turning point (TP), the share of firms with FSTS within the band is 0.6 % (6 of 989) in 2009, 1.2 % (11 of

956) in 2015, 0.9 % (9 of 1,013) in 2023 and 1.0 % (29 of 2,958) pooled. The bulk of mass lies at FSTS = 0; the turning-point estimates are identified primarily through the contrast between non-exporters and exporters and the right tail above the TP, not through dense within-band variation.

Readers should bear this density structure in mind when interpreting the precise

turning-point magnitude.

Robustness to Endogeneity and Selection Three complementary approaches address selection and endogeneity concerns. A note on the selection structure is warranted before presenting the checks. Exporters self-select into foreign markets, so the OLS estimates reflect both selection and treatment effects. As a boundary condition, exporter participation rates declined from 28.4% (2009) to 18.8% (2023) — consistent with servicification of the economy and the gradual exit of marginal exporters — which itself attenuates the threat of a constant positive-selection story across waves. A Heckman two-stage check confirms this reading: Heckman two-step corrections (Panel E) yield insignificant inverse-Mills ratios across all waves (all $|\lambda| < 0.84$, $p > .25$ in each wave), indicating no detectable selection bias on the exporter subsample. The inverse Mills ratio coefficient does not materially change FSTS coefficient estimates across waves, suggesting that selection bias is limited in this context and that the participation-margin productivity gap documented in Section 4.4 reflects a genuine performance discontinuity rather than an artefact of positive selection into exporting. Propensity-score matching (Panel J) corroborates the average H2 TCI association and the baseline DAI_z level association without imposing linearity: ATT estimates for foreign-technology / certification status (0.609–0.637, $p < .001$) and for website ownership (0.298–0.321, $p < .001$) are substantively consistent with the OLS results, though the latter is best read as a descriptive Tier-1 selection-corrected association rather than a capability test. Full matching-balance diagnostics and ATT estimates are reported in Online Appendix Table A.

Two-stage least-squares estimation using leave-one-out sector × region × wave peer-adoption rates as instruments (Panel K; first-stage F-statistics 22–35, well above the Staiger–Stock threshold) returns an instrumented DAI estimate of 0.018 ($p = .942$) and an instrumented TCI estimate of 1.639 ($p < .001$). The null 2SLS DAI coefficient is a key robustness result: it indicates that website presence does not plausibly cause productivity improvements in 2023 Vietnam, reinforcing the Tier 1 proxy obsolescence interpretation of the negative DAI×FSTS interaction. The strongly positive instrumented TCI coefficient (β=1.639, p<.001 — substantially larger than the OLS estimate, consistent with attenuation-bias correction) further indicates that foreign-technology and standards capability is the robust productivity-relevant mechanism in this institutional setting. The instrument set (peer-adoption rates within sector × region × wave cells) satisfies relevance (strong first stage) and approximate exclusion (peer-adoption rates are unlikely to affect individual firm productivity through channels other than DAI adoption, conditional on sector-wave cells). Oster (2019) δ-stability bounds (assuming R_max = 1.3 × R_controlled) indicate that no focal coefficient changes sign or collapses to zero under plausible magnitudes of unobserved selection. Full 2SLS first-stage results and Oster bound calculations are reported in Online Appendix Table B.

Measurement-sensitivity probes indicate that core inferences do not depend on composite construction. Enriching TCI with R&D and product-innovation items (Panel A) attenuates the TCI coefficient modestly but does not alter qualitative conclusions. The enriched DAI_rich composite (Tier 1–2, Panel B) yields directionally consistent moderation in 2023 with weaker significance. Micro-firm exclusion (Panel D) preserves the inverted-U and TCI associations. Paternoster et al. cross-wave z-tests (Panel F) indicate that the DAI cross-wave shift is statistically distinguishable while curvature parameters share a common pooled magnitude. Panel-level estimates are reported in Online Appendix Tables C–F.

Multiple-testing caveat. 4.5 reports four narrative panels (G, H, I, F) and three robustness families (endogeneity/selection, measurement sensitivity) across multiple focal terms. We do not apply a formal multiple-testing correction because the panels probe different identification concerns rather than testing the same hypothesis repeatedly, but readers should weight any single marginal panel result accordingly. Our substantive inferences in Section 5 rely on the pattern across panels and the directional consistency of the focal estimates rather than on the significance of any single robustness panel.

Table 4 collates the robustness panels documented in 4.5 in a single overview to ease crosscomparison. Table LM reports the implied turning points of the inverted-U specification (M2) and the Lind–Mehlum p-values for each wave and the pooled sample.

Table 4: Robustness panels A–K and supplementary checks.

Panel	Sample	N	Term	b (SE)	p
DAI rich DAI rich bin	2023	1013	FSTSc	+1.002† (0.607)	0.099

Panel	Sample	N	Term	b (SE)	p
DAI rich DAI rich bin	2023	1013	FSTSc2	-1.707* (0.813)	0.036
DAI rich DAI rich bin	2023	1013	TCI_z	+0.129** (0.045)	0.004
DAI rich DAI rich bin	2023	1013	DAI_rich_bin_z	-0.009 (0.100)	0.926
DAI rich DAI rich bin	2023	1013	FSTSc_DAI_rich_bin	-0.893 (0.649)	0.169
DAI rich DAI rich bin	2023	1013	FSTSc2_DAI_rich_bin	+1.038 (0.878)	0.237
DAI rich DAI rich bin	2023	1013	joint_F_DAI_rich_interactions	+1.418	0.243
DAI rich DAI rich cont	2023	1013	FSTSc	+1.032† (0.569)	0.070
DAI rich DAI rich cont	2023	1013	FSTSc2	-1.744* (0.757)	0.021
DAI rich DAI rich cont	2023	1013	TCI_z	+0.126** (0.045)	0.005
DAI rich DAI rich cont	2023	1013	DAI_rich_cont_z	-0.004 (0.083)	0.957
DAI rich DAI rich cont	2023	1013	FSTSc_DAI_rich_cont	-0.933† (0.526)	0.076
DAI rich DAI rich cont	2023	1013	FSTSc2_DAI_rich_cont	+1.052 (0.722)	0.145
DAI rich DAI rich cont	2023	1013	joint_F_DAI_rich_interactions	+2.348†	0.099
DAI thin on rich sample 2023	2023	1013	FSTSc	+1.072* (0.493)	0.030

Panel	Sample	N	Term	b (SE)	p
DAI thin on rich sample 2023	2023	1013	FSTSc2	-1.793** (0.666)	0.007
DAI thin on rich sample 2023	2023	1013	TCI_z	+0.129** (0.045)	0.004
DAI thin on rich sample 2023	2023	1013	DAI_z	-0.011 (0.073)	0.880
DAI thin on rich sample 2023	2023	1013	FSTSc_DAIz	-0.912* (0.450)	0.043
DAI thin on rich sample 2023	2023	1013	FSTSc2_DAIz	+1.043† (0.633)	0.100
DAI thin on rich sample 2023	2023	1013	joint_F_DAI_thin_interactions	+2.783†	0.062
IV 2SLS K DAI	pooled	2298	DAI_z (instrumented)	+0.018 (0.249)	0.942
IV 2SLS K TCI	pooled	2298	TCI_z (instrumented)	+1.639*** (0.299)	0.000
PSM J NN1 caliper005	pooled	1085	ATT_DAI_treat	+0.298*** (0.061)	0.000
PSM J NN1 caliper005	pooled	644	ATT_TCI_treat	+0.637*** (0.077)	0.000
PSM J kernel bw006	pooled	1085	ATT_DAI_treat	+0.321*** (0.043)	0.000
PSM J kernel bw006	pooled	639	ATT_TCI_treat	+0.609*** (0.056)	0.000
TCI full direct	2015	956	FSTSc	+1.139* (0.539)	0.035
TCI full direct	2015	956	FSTSc2	-2.088** (0.748)	0.005

Panel	Sample	N	Term	b (SE)	p
TCI full direct	2015	956	TCI_full_z	+0.056 (0.049)	0.253
TCI full direct	2015	956	DAI_z	-0.033 (0.051)	0.520
TCI full direct	2023	1013	FSTSc	+0.675 (0.475)	0.155
TCI full direct	2023	1013	FSTSc2	-1.266* (0.646)	0.050
TCI full direct	2023	1013	TCI_full_z	+0.096* (0.043)	0.024
TCI full direct	2023	1013	DAI_z	+0.097* (0.046)	0.037
TCI full moderation	2015	956	FSTSc	+1.347* (0.598)	0.024
TCI full moderation	2015	956	FSTSc2	-2.350** (0.829)	0.005
TCI full moderation	2015	956	TCI_full_z	+0.004 (0.072)	0.960
TCI full moderation	2015	956	FSTSc_TCIfull	-0.545 (0.495)	0.270
TCI full moderation	2015	956	FSTSc2_TCIfull	+0.641 (0.694)	0.356
TCI full moderation	2015	956	joint_F_TCI_full_interactions	+0.7096	0.492
TCI full moderation	2023	1013	FSTSc	+0.949† (0.574)	0.098
TCI full moderation	2023	1013	FSTSc2	-1.579* (0.777)	0.042
TCI full moderation	2023	1013	TCI_full_z	+0.117* (0.049)	0.016
TCI full moderation	2023	1013	FSTSc_TCIfull	-0.317 (0.314)	0.314
TCI full moderation	2023	1013	FSTSc2_TCIfull	+0.185 (0.431)	0.668
TCI full moderation	2023	1013	joint_F_TCI_full_interactions	+1.8868	0.152

Panel	Sample	N	Term	b (SE)	p
common N reconciled 2023	M8	1013	FSTSc	+1.072* (0.493)	0.030
common N reconciled 2023	M8	1013	FSTSc2	-1.793** (0.666)	0.007
common N reconciled 2023	M8	1013	TCI_z	+0.129** (0.045)	0.004
common N reconciled 2023	M8	1013	DAI_z	-0.011 (0.073)	0.880
common N reconciled 2023	M8	1013	FSTSc_DAIz	-0.912* (0.450)	0.043
common N reconciled 2023	M8	1013	FSTSc2_DAIz	+1.043† (0.633)	0.100
common N reconciled 2023	2023	1013	joint_F_DAI_M8	+2.7833†	0.062
exporter only	M2	281	FSTSc	-0.611* (0.242)	0.012
exporter only	M2	281	FSTSc2	+1.674* (0.853)	0.050
exporter only	M7	281	FSTSc	-0.409† (0.238)	0.085
exporter only	M7	281	FSTSc2	+1.585† (0.854)	0.064
exporter only	M7	281	TCI_z	+0.149† (0.077)	0.053
exporter only	M7	281	DAI_z	+0.192* (0.090)	0.034
exporter only	M8	281	FSTSc	-0.403 (0.262)	0.124
exporter only	M8	281	FSTSc2	+1.377 (0.918)	0.134

Panel	Sample	N	Term	b (SE)	p
exporter only	M8	281	TCI_z	+0.147† (0.077)	0.057
exporter only	M8	281	DAI_z	+0.075 (0.143)	0.598
exporter only	M8	281	FSTSc_DAIz	+0.078 (0.238)	0.743
exporter only	M8	281	FSTSc2_DAIz	+0.802 (0.874)	0.359
exporter only	2009_exp	281	joint_F_DAI_M8	+0.4235	0.655
exporter only	M2	198	FSTSc	-0.773** (0.290)	0.008
exporter only	M2	198	FSTSc2	-2.204* (0.957)	0.021
exporter only	M7	198	FSTSc	-0.771** (0.296)	0.009
exporter only	M7	198	FSTSc2	-2.619** (0.957)	0.006
exporter only	M7	198	TCI_z	+0.242*** (0.071)	0.001
exporter only	M7	198	DAI_z	-0.212† (0.112)	0.059
exporter only	M8	198	FSTSc	-0.672* (0.301)	0.025
exporter only	M8	198	FSTSc2	-2.719* (1.147)	0.018
exporter only	M8	198	TCI_z	+0.237*** (0.071)	0.001
exporter only	M8	198	DAI_z	-0.177 (0.202)	0.383
exporter only	M8	198	FSTSc_DAIz	-0.335 (0.277)	0.226
exporter only	M8	198	FSTSc2_DAIz	-0.172 (1.096)	0.876
exporter only	2015_exp	198	joint_F_DAI_M8	+0.7344	0.481

Panel	Sample	N	Term	b (SE)	p
exporter only	M2	190	FSTSc	-0.910** (0.335)	0.007
exporter only	M2	190	FSTSc2	-1.461 (1.160)	0.208
exporter only	M7	190	FSTSc	-0.856* (0.344)	0.013
exporter only	M7	190	FSTSc2	-1.397 (1.171)	0.233
exporter only	M7	190	TCI_z	+0.061 (0.067)	0.366
exporter only	M7	190	DAI_z	+0.037 (0.086)	0.665
exporter only	M8	190	FSTSc	-1.079** (0.332)	0.001
exporter only	M8	190	FSTSc2	-2.921** (1.094)	0.008
exporter only	M8	190	TCI_z	+0.080 (0.067)	0.234
exporter only	M8	190	DAI_z	-0.271† (0.141)	0.054
exporter only	M8	190	FSTSc_DAIz	+0.510† (0.294)	0.083
exporter only	M8	190	FSTSc2_DAIz	+2.676** (1.021)	0.009
exporter only	2023_exp	190	joint_F_DAI_M8	3.4455*	0.034
exporter only	M2	669	FSTSc	-0.861*** (0.167)	0.000
exporter only	M2	669	FSTSc2	-0.200 (0.581)	0.730
exporter only	M7	669	FSTSc	-0.704*** (0.173)	0.000
exporter only	M7	669	FSTSc2	-0.172 (0.577)	0.765
exporter only	M7	669	TCI_z	+0.178*** (0.042)	0.000

Panel	Sample	N	Term	b (SE)	p
exporter only	M7	669	DAI_z	+0.066 (0.057)	0.247
exporter only	M8	669	FSTSc	-0.683*** (0.183)	0.000
exporter only	M8	669	FSTSc2	-0.497 (0.643)	0.440
exporter only	M8	669	TCI_z	+0.179*** (0.042)	0.000
exporter only	M8	669	DAI_z	-0.025 (0.104)	0.809
exporter only	M8	669	FSTSc_DAIz	-0.001 (0.164)	0.995
exporter only	M8	669	FSTSc2_DAIz	+0.702 (0.625)	0.261
exporter only	pooled_exp	669	joint_F_DAI_M8	+0.7733	0.462
micro excluded pooled	pooled_l1ge10	2473	FSTSc	+0.794** (0.301)	0.008
micro excluded pooled	pooled_l1ge10	2473	FSTSc2	-1.647*** (0.434)	0.000
micro excluded pooled	pooled_l1ge10	2473	TCI_z	+0.188*** (0.028)	0.000
micro excluded pooled	pooled_l1ge10	2473	DAI_z	+0.054 (0.051)	0.288
micro excluded pooled	pooled_l1ge10	2473	FSTSc_DAIz	-0.387 (0.281)	0.169
micro excluded pooled	pooled_l1ge10	2473	FSTSc2_DAIz	+0.405 (0.416)	0.330
micro excluded pooled	pooled_l1ge10	2473	joint_F_DAI_interactions	+1.7902	0.167

Panel	Sample	N	Term	b (SE)	p
sector split manufacturing	M2	1854	FSTSc	+0.971** (0.296)	0.001
sector split manufacturing	M2	1854	FSTSc2	-1.883*** (0.426)	0.000
sector split manufacturing	M7	1854	FSTSc	+0.662* (0.296)	0.025
sector split manufacturing	M7	1854	FSTSc2	-1.382** (0.428)	0.001
sector split manufacturing	M7	1854	TCI_z	+0.223*** (0.033)	0.000
sector split manufacturing	M7	1854	DAI_z	+0.087** (0.033)	0.009
sector split manufacturing	M8	1854	FSTSc	+0.840* (0.328)	0.010
sector split manufacturing	M8	1854	FSTSc2	-1.624*** (0.473)	0.001
sector split manufacturing	M8	1854	TCI_z	+0.228*** (0.033)	0.000
sector split manufacturing	M8	1854	DAI_z	+0.034 (0.057)	0.552
sector split manufacturing	M8	1854	FSTSc_DAIz	-0.543† (0.310)	0.079
sector split manufacturing	M8	1854	FSTSc2_DAIz	+0.626 (0.457)	0.170

Panel	Sample	N	Term	b (SE)	p
sector split manufacturing	pooled	1854	joint_F_DAI_interactions_M8	+2.286	0.102
sector split manufacturing	pooled	1854	joint_F_TCI_interactions_M3	+4.571*	0.011
sector split non manufacturing	M2	1104	FSTSc	+1.615† (0.870)	0.064
sector split non manufacturing	M2	1104	FSTSc2	-2.479* (1.244)	0.046
sector split non manufacturing	M7	1104	FSTSc	+1.350 (0.890)	0.130
sector split non manufacturing	M7	1104	FSTSc2	-2.039 (1.275)	0.110
sector split non manufacturing	M7	1104	TCI_z	+0.090† (0.054)	0.096
sector split non manufacturing	M7	1104	DAI_z	+0.068 (0.045)	0.133
sector split non manufacturing	M8	1104	FSTSc	+1.387 (0.868)	0.110
sector split non manufacturing	M8	1104	FSTSc2	-2.115† (1.237)	0.087
sector split non manufacturing	M8	1104	TCI_z	+0.091† (0.054)	0.091
sector split non manufacturing	M8	1104	DAI_z	+0.060 (0.122)	0.622

Panel	Sample	N	Term	b (SE)	p
sector split non manufacturing	M8	1104	FSTSc_DAIz	-0.190 (0.719)	0.791
sector split non manufacturing	M8	1104	FSTSc2_DAIz	-0.341 (1.065)	0.749
sector split non manufacturing	pooled	1104	joint_F_DAI_interactions_M8	+1.2729	0.280
sector split non manufacturing	pooled	1104	joint_F_TCI_interactions_M3	+4.9441**	0.007
wave interaction pooled	pooled	2958	joint_F_FSTSc_x_wave	+0.8901	0.411
wave interaction pooled	pooled	2958	joint_F_FSTSc2_x_wave	+0.7756	0.461
wave interaction pooled	pooled	2958	joint_F_DAIz_x_wave	+4.1656*	0.016
wave interaction pooled	pooled	2958	joint_F_TCIz_x_wave	+1.1156	0.328
wave interaction pooled	pooled	2958	joint_F_all_wave_interactions	+2.0664*	0.036

Notes. Robustness panels: TCI_full = broader TCI composite; DAI_rich = richer DAI composite (2023 only). HC1 robust standard errors. † $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

Panel / sample N Focal coefficient Joint test A. TCI_full direct (2015) 956 TCI_full_z = 0.056 (p = .253) n.s.; DAI_z = M3 joint p = .492 n.s. −0.033 n.s. A. TCI_full direct (2023) 1,013 TCI_full_z = 0.096 (p = .024); DAI_z = M3 joint p = .152 n.s. 0.097 (p = .037) B. DAI_rich continuous (2023) 1,013 FSTS_c × DAI_rich_cont_z = −0.933 † (p M8 joint p = .099 † = .076) B. DAI_rich binary (2023) 1,013 FSTS_c × DAI_rich_bin_z

= −0.893 (p = M8 joint p = .243 n.s. .169) n.s. C. Common-N reconciled (2023) 1,013 DAI_z moderation re-estimated on identical M8 joint p = .062 † 2023 sample D. Micro-firm exclusion (l1 ≥ 10) 2,473 TCI_z = 0.188 ; DAI_z = 0.054 n.s. M8 joint p = .167 n.s. G. Manufacturing (sector1 1,854 TCI_z = 0.223 ; DAI_z = 0.087 ; M8 joint p = .103 † {1, 2, 3}) FSTS_c × DAI_z = −0.543 † H. Exporter-only (FSTS > 0, 669 FSTS_c = −0.861 ; FSTS_c² = −0.200 M8 joint p = .462 n.s. pooled) n.s. I. Wave × focal interaction 2,958 DAI_z × wave detectable (p = .016); other Only DAI direct shifts (pooled) focal × wave n.s. cross-wave-separable Density around TP (±5 pp band) 29 / 2,958 1.0% of firms within band; bulk of mass at Curvature identified by parFSTS = 0 ticipation margin J. PSM ATT — website (1-NN, 1,085 ATT_DAI = 0.298 (SE 0.061) Positive and large under caliper 0.05) matching J. PSM ATT — website (kernel 1,085 ATT_DAI = 0.321 (SE 0.043) Robust to algorithm BW 0.06) J. PSM ATT — cert / foreign-tech 640 ATT_TCI = 0.637 (SE 0.078) Positive and large under (1-NN) matching K. IV / 2SLS — DAI_z 2,298 DAI_z = 0.018 (p = .942) n.s. First-stage F = 34.6 (strong) K. IV / 2SLS — TCI_z 2,298 TCI_z = 1.639 (SE 0.299) First-stage F = 22.1 (strong) Oster (2019) δ = 1 bounds 2,958 FSTSc 0.92 to 0.61; FSTSc2 −2.21 to −1.17; No sign change; focal findTCI 0.14 to 0.19; DAI 0.07 to 0.08 ings stable

Each row summarises one robustness panel from 4.5 estimated by OLS with HC1 robust standard errors (PSM and 2SLS use the indicated alternative estimators). Significance markers: p < .001, p < .01, p < .05, † p < .10, n.s. = not significant. Panel E (Heckman / control function selection corrections) and Panel F (Paternoster cross-wave z-tests, Paternoster et al., 1998) are reported in the prose because their natural presentation is per-sample rather than per-row. Source: World Bank Enterprise Surveys (Vietnam 2009, 2015, 2023), www.enterprisesurveys.org; authors' calculations. Notes.

Table 5: Implied turning points of the inverted-U (M2 specification).

Sample	Turning point (FSTS)	95% CI	Lind-Mehlum p	FSTS range
2009 (N=989)	46.2%	[37.4%, 55.1%]	0.006**	[0%, 100%]
2015 (N=956)	39.3%	[30.3%, 48.4%]	0.009**	[0%, 100%]
2023 (N=1,013)	41.6%	[31.7%, 51.5%]	0.013*	[0%, 100%]
Pooled (N=2,958)	39.7%	[34.0%, 45.5%]	0.000***	[0%, 100%]

Notes. Turning points estimated from M2 (quadratic FSTS specification). Lind-Mehlum (2010) test p-value for inverted-U shape. 95% CI computed by delta method. All four samples confirm an inverted-U with turning point in the 39–46% FSTS range.

Paternoster (1998) cross-wave coefficient stability tests (M7):

Comparison	FSTSc z (p)	FSTSc² z (p)	TCI_z z (p)	DAI_z z (p)
2009 vs. 2015	z=-0.61 (p=0.543)	z=0.96 (p=0.337)	z=1.29 (p=0.198)	z=3.35 (p=0.001)
2009 vs. 2023	z=0.09 (p=0.926)	z=0.14 (p=0.888)	z=1.42 (p=0.155)	z=1.28 (p=0.201)
2015 vs. 2023	z=0.66 (p=0.509)	z=-0.84 (p=0.400)	z=0.07 (p=0.947)	z=-2.05 (p=0.040)

Notes. Cross-wave coefficient equality tests following Paternoster et al. (1998). Failure to reject indicates coefficient stability across waves. FSTSc and FSTSc² comparisons support wave-specific heterogeneity in TCI and DAI direct effects but not in the I–P curvature parameters. Turning points (already reported in Table 5 above) are derived from M2 ($\ln LP = \beta_0 + \beta_1$ FSTS_c + β_2 FSTS_c² + controls + sector FE [+ wave FE in pooled]) back-transformed to the raw FSTS scale; 95% CI uses the delta method; Lind–Mehlum p-values follow the Sasabuchi-style endpoint test (Lind & Mehlum, 2010). Source: WBES Vietnam 2009, 2015, 2023; authors' calculations.

5. Discussion

The results are best located first against the meta-analytic baseline. Wu, Fan, and Chen (2022, MIR) establish a positive average I–P relationship for emerging-market multinationals — a useful macro-level anchor, though one that aggregates across institutional contexts in which the functional form may vary substantially. That meta-analysis leaves country-level heterogeneity in the I–P curve shape under-specified. Vietnam, a single-party transitional economy with rapid but uneven digital infrastructure uptake, offers a critical test of whether the pooled meta-level estimate masks functional-form nonlinearity at the within-country level. The present evidence speaks to that concern: the inverted-U is robust within Vietnam, yet a participation-margin step drives it rather than within-exporter saturation — a structure that pooling exporters and non-exporters in cross-country meta-analyses tends to inflate into an apparent continuous curvature. The within-Vietnam temporal durability of the threshold (39–46 % band sustained across 14 years) additionally indicates that institutional context — held constant in this single-country design — may stabilise the curve shape in a way that cross-country designs cannot isolate. Our finding of a reproducible inverted-U at FSTS ≈ 39–46% in Vietnam, despite the cross-national skepticism articulated by Pisani, Garcia-Bernardo, and Heemskerk (2020, SMJ), is consistent with the view that single-country institutional contexts retain functional-form heterogeneity that pooled cross-national samples obscure through aggregation. The ascending-limb mechanism receives independent corroboration from matched firm-level customs panel evidence: Vietnamese firms that absorbed a positive US demand shock over 2018–2020 expanded exports to both US and non-US destinations and recorded commensurate gains in labour productivity, with demand-driven export expansion explaining 68.5 per cent of the observed domestic value-added ratio increase over the same period (Agarwal, Barattieri, & Mattoo, 2026). This corroboration strengthens the interpretation that moderate export intensity generates genuine productivity gains in Vietnamese firms — not merely a compositional artefact of positive selection into exporting.

5.1 Reinterpreting the baseline website-presence indicator in Vietnam

The descriptive DAI_z findings carry one central implication: baseline website presence in Vietnam should not be read as a universal, temporally stable productivity premium, nor as evidence on dynamic digital capability. TCI_z and DAI_z are both positive on average in the pooled specifications, yet their empirical roles diverge materially across waves and across identification strategies (Vahlne, 2020; Stallkamp and Schotter, 2021). Within the limits of a Tier-1 binary website indicator, what emerges is that website ownership is no constant background advantage; it is a context-sensitive marker whose productivity association attenuates under instrumental-variable identification and varies sharply across the 2009 / 2015 / 2023 waves.

The negative sign on the FSTS_c × DAI_z interaction in 2023 — where DAI productivity relevance attenuates at higher export intensities — is consistent with the Tier-1 boundary of the construct: a website alone cannot manage the transaction density of high-export-intensity operations without electronic-payment and process-integration support. This contrasts with settings where the same construct includes Tier 2 transaction-enabling items, where higher export intensity could instead amplify the productivity return to digital adoption. This reading helps reconcile the coexistence of positive pooled effects with uneven wave-specific results. The pooled model captures the average tendency for stronger capability to track better performance; the wave-specific models show that this tendency is not equally strong in every phase. The value of digital capability thus depends on where a firm stands in the broader trajectory of internationalisation and transition — as seen in repeated cross-sectional snapshots rather than within-firm longitudinal records.

5.2 Why the distinction between TCI and baseline DAI matters

The results strengthen the methodological case for keeping foreign-technology / standards capability apart from a baseline website-presence indicator, rather than folding the two into a single "digital adoption" or "digital transformation" label. The PSM and IV evidence in 4.5 (Panels J and K) makes the distinction sharper (Karna et al., 2016).

TCI is robust under both matching and

instrumentation: the 2SLS estimate of TCI_z is 1.64 ($p < .001$) under a strong instrument (first-stage $F = 22.1$), and the matching ATT for the cert / foreign-tech treatment is 0.61–0.64 ($p < .001$) on a matched sample of approximately 640 firms.

By contrast, the OLS-detected

DAI direct association is reproduced under PSM (matching ATT = 0.30–0.32, $p < .001$) but attenuates to a null under 2SLS ($\beta = 0.02$, $p = .94$). Taken together, the results suggest that foreign-technology / standards capability behaves like a more identification-robust productivity channel, whereas website-based digital presence behaves like a more context-sensitive and selection-sensitive marker of performance heterogeneity. The DAI_rich extension reported in 4.5 Panel B reinforces the construct interpretation rather than weakening it. Although the primary DAI_z anchored on c22b (website presence) is by 2023 close to a Tier-1 baseline indicator, the DAI_rich extension available only in 2023 — combining c22b with electronic-payment shares (k33, k38) — produces a similarly directioned and a moderation pattern in the same direction (FSTS_c × DAI_rich_cont_z = -0.93, M8 joint $p = .099$). Whether digital adoption is measured by the thin Tier-1 website indicator or by the deeper Tier-2 / 3 transaction items, the 2023 moderation pattern goes in the same direction. This common-direction evidence guards against the proxy-obsolescence reading of the DAI pattern in isolation: even under a richer measurement that is harder to dismiss as a routine marker, the within-2023 moderation is consistent with the headline finding. The difference between TCI and DAI matters because the two constructs do not behave identically across periods or under different identification strategies.

TCI_z behaves like a

robust foreign-technology / standards capability whose productivity association survives both PSM and 2SLS. DAI_z behaves like a context-sensitive marker whose productivity association is documented under matched comparison but not under exogenous instrumental variation. This

argues for a more careful treatment of digitalisation in international-business research — one that distinguishes basic digital enablement (easily contaminated by selection on observables) from broader technological depth (more identification-robust), rather than collapsing them into a single label.

5.3 The significance of the 2015 dip

Before invoking the macro-infrastructure narrative, the 2015 null should be grounded in the micro-structure of the sample itself. As reported in Table 2, the exporter share fell from 28.4% in 2009 to 20.7% in 2015 — a selective exit that left a residual cohort disproportionately concentrated in labour-intensive, OEM-dependent assembly operations. For a garment or electronics sub-contractor awaiting orders from a foreign principal, a basic corporate website functions as a digital pamphlet rather than a productivity-enabling coordination tool: it signals presence but does not optimise supply-chain flows, enable transaction-level data integration, or substitute for the digital interfaces controlled by the upstream buyer. This cohort-composition mechanism — not merely the macro infrastructure gap — helps explain why the website-based DAI coefficient compresses to a null in 2015 even as the inverted-U curvature remains statistically present. The proxy-obsolescence channel reinforces this reading: by 2015, website ownership had diffused sufficiently that the c22b indicator could no longer separate digitally capable exporters from routine web presence, further attenuating its measured association with productivity. We therefore treat the macro indicators below as institutional context that makes the wave-specific reading plausible, not as identifying variation.

A purely stage-contingency interpretation is not the only reading of the wave-specific DAI pattern. An alternative — and to a degree complementary — explanation is proxy obsolescence: the c22b indicator of website ownership measures something different in 2009, 2015 and 2023 even though the question text is identical across waves.

In 2009, having a corporate website

was a relatively distinguishing market interface that signalled basic digital legitimacy in international transactions; by 2023, a website is closer to a routine marker of basic digital presence shared across exporting and non-exporting firms alike. Under this reading, the falling DAI direct association across waves partly reflects the shrinking informational content of c22b rather than a structural change in the productivity payoff to foundational digital adoption itself. We do not view proxy obsolescence and stage contingency as mutually exclusive: both can be operating jointly. We flag the proxy-obsolescence reading because it is not testable within the cross-wavecomparable WBES instrument and future work using waves with richer Tier 2–3 digital items is needed to disentangle the two channels.

The 2015 pattern is best read not as an anomaly, but as a wave-specific compression of the foundational digital-adoption channel under transitional infrastructure conditions. Even where the nonlinear I–P structure becomes clearer, the direct payoff from website-based digital presence can compress to a null. This reading is institutionally plausible, but it should not be treated as a formally identified explanation of the coefficient pattern. Firms may pass through a transitional stage in which export expansion remains important but the productivity contribution of basic digital adoption becomes more difficult to realise or detect within-sample. We treat the 2015 compression honestly as a wave-specific association consistent with stage contingency rather than as a fully cross-wave-identified structural shift. As reported in 4.5 Panel I, the formal pooled wave × focal interaction test detects only the DAI direct shifts as cross-wavedistinguishable; the FSTS curvature and the FSTS × DAI moderation cross-wave differences are not statistically separable.

The macro context of the period is consistent with a digital-

infrastructure trough in 2015 relative to the 2009 and 2023 anchors. Public secondary indicators support this reading.

World Bank World Development Indicators (WDI) record Vietnam's

individuals-using-the-Internet share at approximately 26 % in 2009, 45 % in 2015 and over 78 % by 2023; ITU's fixed-broadband subscriptions per 100 people grew from approximately 3 in 2009 to 8 in 2015 and to over 20 by 2023.

Vietnam's National Digital Transformation

Programme (NDTP) was issued in 2020, and cross-border e-payment platforms (VNPAY, MoMo, ZaloPay) and B2B exporter marketplaces (Alibaba.com, Amazon Global Selling) reached scale in Vietnam only after 2018–2019. The 2015 wave therefore observes Vietnamese exporters during a transitional infrastructure phase in which a website could not yet plug into a transactionsupporting digital ecosystem; the 2023 wave observes them after the post-NDTP scaffolding had matured. We do not treat these macro indicators as identifying variation — they are not entered into the regression — but they make the wave-specific reading institutionally plausible rather than purely post-hoc. Rather than treating this wave as an anomaly, it is more useful to interpret it as evidence of wave-specific heterogeneity that is consistent with stage contingency. The dip shows why pooled averages alone are insufficient. Without the wave-specific analysis, one would miss the possibility that capability payoffs compress or fade temporarily before re-emerging in a later phase, even when the curvature parameters of the I–P relationship themselves remain statistically indistinguishable across waves.

5.4 Managerial implications

The managerial implication is not that firms should invest less in digitalisation. Rather, it is that the *composition* of capability investment should change with the firm's position relative to two empirically anchored thresholds: the export-participation barrier (FSTS = 0 to FSTS > 0) and the inverted-U turning-point band at 39–46 % FSTS. Read together, the step-function structure at the participation margin and the Tier-1 saturation of website-based digital presence imply a threshold-conditioned capital-allocation map for Vietnamese SME managers — one that translates the wave-specific coefficient patterns into stage-appropriate priorities.

Stage 1 — Pre-export and low-intensity firms (FSTS = 0 or FSTS < ~30 %). The dominant productivity-relevant action at this stage is crossing the participation barrier, not optimising digital presence per se. Foundational website adoption and basic e-payment connectivity remain useful as low-cost legitimacy signals to potential foreign buyers, but their productivity contribution is bounded by the Tier-1 ceiling of the WBES instrument and by the diffusion-driven attenuation of c22b informational content documented in the descriptive evidence. Capital expenditure on incremental website redesign or marketing-digital extensions at this stage delivers limited marginal productivity return relative to investment that loosens the fixed-cost barriers to first export participation: foreign-market intelligence, ISO-style quality certification, and entry-cost financing.

Stage 2 — Approaching the turning-point band (FSTS ≈ 30–46 %). The TCI direct association (pooled $\beta = 0.179$, $p < .001$; instrumented $\beta = 1.639$) and the curvature-flattening role of TCI moderation (M3 joint $p = .003$ pooled) imply that the binding productivity friction in this band is the coordination ceiling that foreign-technology and standards capability is most effective at relaxing. Capital expenditure should therefore tilt toward TCI-deepening investments — foreign-licensed technology, formal quality-certification systems, process integration with foreign principals — rather than incremental DAI upgrades on the Tier-1 layer. The basic website is by this stage a hygiene factor; what differentiates productivity outcomes within the cohort is whether the firm has absorbed the deeper technological and standards capability that absorbs coordination cost at higher export intensity.

Stage 3 — Beyond the turning point (FSTS > ~46 %). The negative FSTS × DAI_z interaction observed in 2023 ($\beta = -0.912$, $p = .043$) is consistent with the Tier-1 boundary of the WBES website indicator and with the population-diffusion attenuation of c22b informational content. For high-intensity exporters, marginal investment in Tier-1 website infrastructure delivers a negative incremental productivity association; the binding friction is the absence of Tier 2–3 process-integration, payment, and supply-chain digitisation, which the WBES Tier-1 instrument cannot capture but which the DAI_rich extension (Panel B) and the qualitative cohort evidence both support as the relevant upgrade path.

The *2015 dip* in DAI direct association is interpretable through this lens as well. The exporter cohort in 2015 was disproportionately concentrated in OEM-dependent assembly operations where a corporate website functioned as a digital pamphlet rather than as a coordination tool — a setting in which Tier-1 digital infrastructure cannot substitute for the upstream-buyer-controlled digital interfaces that mediate the actual transaction flow. For managers of OEM-style firms, the implication is that pure DAI upgrades do not loosen the productivity ceiling when the binding constraint is upstream digital governance; the productivity-relevant capability lever in such cohorts is upgrading toward TCI-anchored channels that improve the firm's standing relative to foreign principals.

Two cross-cutting principles emerge. First, managers should avoid conflating basic digital adoption with deeper technological capability — the constructs do not behave identically across periods or under identification-robust estimation, and Tier-1 digital expenditure does not substitute for the foreign-technology, standards, and process-integration investments that the TCI channel reflects. Second, allocation decisions should be staged against the empirically identified thresholds rather than against generic digital-transformation prescriptions: investment in incremental Tier-1 presence at high export intensity may absorb scarce capital with limited productivity return, while investment in TCI-channel capability around the 39–46 % band corresponds to where the cross-wave evidence is most consistent with a productivity-enhancing effect. The threshold-conditioned framing translates the step-function and Tier-1 saturation findings into actionable allocation guidance without overstating the causal status of the cross-sectional coefficients.

5.5 Policy implications

The policy reading here is offered as tentative consideration, not directive prescription. The associational nature of the evidence, the breadth of the wave-specific heterogeneity, and the single-economy scope all weigh against converting these findings into firm policy targets. With those caveats kept explicit, three considerations follow for Vietnam's trade and digital-economy policy design. First, export-promotion instruments designed around a uniformly positive internationalisation premium will overshoot in transitional periods such as the 2015 wave, when capability payoffs were compressed even as export intensity remained productivity-relevant. Targeting export support at firms positioned in the entry-cost zone, rather than at firms already operating at high export intensity, is consistent with the curvature documented in the pooled and later-wave samples. Second, digital-economy programmes that treat foundational adoption (websites, basic e-payment) as a sufficient policy lever are likely to be attenuated by the implementation lag and by the conditional nature of the digital channel at higher export intensity.

Programmes

that bundle Tier 1–2 digital adoption with deeper capability upgrading — quality certification, absorptive-capacity investment, organisational routines for cross-border coordination — are more consistent with the pattern that emerges in 2023. Third, the cross-wave comparison — which, given the repeated cross-sectional design, reflects cohort-level snapshots rather than within-firm trajectories — suggests that policy evaluation windows matter.

A digital-

transformation programme assessed only against a 2015-style transitional baseline would un-

derstate its long-run productivity contribution; a programme assessed against a 2009-style or 2023-style baseline would overstate it relative to the in-between phase. Policy evaluation that takes the wave-specific structural differences seriously will couple short-window outcome measurement with sustained measurement of the capability and infrastructure environment in which firms operate. In the Vietnamese context, the World Bank (2025b) *Taking Stock* update identifies the cultivation of high-tech industrial talent as a binding constraint mediating digital and capability investment into sustainable productivity gains — reinforcing the argument that policy evaluation in transitional settings must monitor the capability environment, not adoption rates alone. At the macro level, Vietnam's service trade liberalization over 2008–2016 raised service-sector productivity by an average of 2.9 per cent per year and manufacturing productivity by 3.1 per cent per year (Barattieri, Mattoo, & Signoret, 2026), suggesting that the ascending limb documented in the WBES microdata reflects in part the economy-wide gains from trade opening that firm-level capability investments convert into sustained productivity advantage.

6. Limitations and future research

The findings should be read against five limitations. The first and most fundamental is that the WBES microdata are repeated cross-sections, not a true firm panel. This design cannot identify within-firm trajectories over time, nor net out time-invariant unobserved heterogeneity at the firm level, so causal attribution for any estimated coefficient remains out of reach. The associational language used throughout the paper reflects this constraint and should not be relaxed in any reader's interpretation of the results. The appropriate analytical remedy is a matched-firm longitudinal panel combining WBES waves with administrative registry data or customs records, which would allow difference-in-differences or fixed-effects identification of the participation-margin productivity jump; Wooldridge (2010) provides the relevant identification framework for such designs.

Second, DAI_z captures only a foundational, Tier-1-style layer of digital adoption centred on website presence, rather than digitally integrated organisational capability. This means the present analysis cannot conclude whether deeper digital integration — Tier 2 electronic-payment intensity, Tier 3 ERP-linked supply-chain digitisation, or Tier 4 data-driven operations — exhibits a more stable or differently shaped productivity channel than the website-presence binary documents here. The appropriate remedy is a panel of digitally integrated firms measured with Tier 2–4 indicators (electronic-payment intensity k33/k38, ERP adoption, platform integration), applying the construct-tier hierarchy of Verhoef et al. (2021) to assign each indicator to its correct layer before entering it in the I–P model.

Third, the analysis is conducted on a single transitional economy. Vietnam is informative precisely because its institutional and digital environment shifted noticeably across the 2009– 2023 observation window, but the cross-wave pattern documented here may not generalise without modification to economies whose digital infrastructure or export composition follows a different trajectory.

Fourth, the cross-wave evidence is uneven in statistical strength. The Paternoster et al. (1998) z-tests indicate that the drop in DAI_z between 2009 and 2015 ($z = 3.353$, $p < .001$) and its recovery between 2015 and 2023 ($z = -2.051$, $p = .040$) are statistically distinguishable, but most other cross-wave coefficient differences sit at marginal or non-significant magnitudes. The cross-wave comparison therefore rests primarily on the directional consistency of the wave-specific estimates and on the concentration of the digital signal in 2023, rather than on uniformly significant pairwise coefficient differences across all focal terms. Sharper identification of the digital channel is achievable by exploiting the exogenous timing of Vietnam's National Digital Transformation Programme (launched June 2020; Prime Minister of Vietnam, 2020) as a policy instrument in a difference-in-differences design that contrasts pre- and post-NDTP firm cohorts within the same WBES sampling frame.

Fifth, the sector fixed effects in the main models are intentionally broad to keep the cross-wave specification comparable. Section 4.5 Panel G addresses one natural extension by re-estimating the pooled M2 / M7 / M8 specifications on manufacturing (sector1 {1, 2, 3}; N = 1,854) versus non-manufacturing (sector1 {4, 5, 6, 7}; N = 1,104) subsets, and finds that the digital-adoption channel operates primarily in manufacturing while the technological-capability channel is present in both. Finer industry-level mechanisms — for example, contrasting digitally intensive services with traditional services, or contrasting export-oriented with domestically oriented manufacturing sub-sectors — would benefit from a research design that exploits richer industry classification than the WBES one-digit codes allow.

Sixth, the analysis is limited to Vietnam as a single transitional economy at a particular stage of its digital and institutional development. Understanding how the DAI moderation pattern documented here — negative at high export intensity under Tier-1-only adoption — compares with patterns in other emerging markets at different institutional stages, or with patterns in digitally advanced economies where the same interaction is predicted to be positive, requires cross-setting comparative designs that systematically vary institutional transaction costs and digital infrastructure quality. Such designs would also allow more direct tests of the context-contingent framing developed in Section 2.1.

7. Conclusion

This study revisits the internationalisation–performance relationship in Vietnam, treating foreign-technology / standards capability as the primary capability construct and retaining a single-item website-presence indicator only as a baseline digital-presence control. Comparing pooled with wave-specific evidence across the 2009 / 2015 / 2023 WBES waves lets it weigh the structural durability of the I–P threshold against the wave-level identification robustness of each capability dimension.

The findings indicate that the internationalisation–performance relationship is non-monotonic in the full sample, that a participation-and-intensity structure drives the pattern rather than strong within-exporter curvature alone, and — critically — that the inverted-U turning point is structurally durable. The threshold range clusters within a narrow ~7-percentage-point band (39.3–46.2 %) across 14 years of major institutional change: WTO accession, the Global Financial Crisis, the COVID-19 supply-chain disruption, and the diffusion of digital-payment infrastructure. Such durability points to deep firm-level constraints — coordination capacity, contract enforcement, and trade-finance access — that move more slowly than the wider operating environment.

The results also suggest that technological capability and the baseline website-presence indicator are not interchangeable, and that conflating them under a single "digital adoption" or "digital transformation" label would obscure their differing identification properties. Technological capability is comparatively stable and identification-robust across specifications; the baseline DAI_z control is more wave-sensitive and attenuates to a null under instrumental-variable estimation, consistent with Tier-1 proxy obsolescence as website ownership diffuses to near-universal levels in the Vietnamese firm population. Within the constraint of a single binary website indicator, this paper makes no claim about dynamic digital capability moderation; any such test awaits richer Tier 2–4 indicators that are not currently available cross-wave in the WBES Vietnam instrument.

Conflict of interest

The authors declare no conflict of interest.

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Data availability statement

The data that support the findings of this study are from the World Bank Enterprise Surveys and are available from the World Bank Enterprise Surveys portal, subject to registration and compliance with the World Bank Enterprise Surveys Data Access Protocol. Because the protocol restricts redistribution of the original .dta files to third parties, the authors do not redistribute the raw survey files. Replication materials, including variable-construction details, model specifications, and computational outputs, are available from the authors upon reasonable request and may be shared to the extent permitted by the data-access terms.

Use of generative AI in the writing process

Generative AI tools were used during manuscript preparation to assist with language editing, structure suggestions, and the assembly of the replication package documentation.

All con-

ceptual framing, hypothesis development, empirical analysis, results interpretation, and final wording were authored by the human authors, who take full responsibility for the content of the publication.

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The replication

package accompanying this manuscript was developed to support methodological transparency and computational reproducibility.

Supplementary materials

All figures and tables in the manuscript are embedded inline. The underlying coefficient outputs are also distributed as plain CSV files in the replication package for downstream reanalysis:

tables/table_1_descriptives.csv (Table 2), tables/coefs_main_models.csv (M0–M8

long-format coefficient table), tables/joint_tests_main_models.csv (H2 and P1 joint F-tests), tables/table_lind_mehlum.csv (Table LM source), tables/table_3_robustness.csv (Table 4 panels A–D + G), tables/selection_checks.csv (Heckman / control function), and tables/table_paternoster.csv (cross-wave z-tests). Replication package and instructions: see p3_vietnam/README.md.

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